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Avascular Growth, Angiogenesis and Vascular Growth in Solid Tumours: The Mathematical Modelling of the Stages of Tumour Development

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Abstract—The growth and development of solid tumours occurs in two distinct stages—the avascular growth phase and the vascular growth phase. During the former growth phase the tumour remains in a diffusion-limited, dormant state of a few millimetres in diameter (cf. multicell spheroids, carcinoma *in situ*) while during the latter growth phase, invasion and metastasis may take place. In order to accomplish the transition from avascular to vascular growth, solid tumours may secrete diffusible substances known as tumour angiogenesis factors (TAF) into the surrounding tissue. Endothelial cells which form the lining of neighbouring blood vessels respond to this chemotactic stimulus in a well-ordered sequence of events. Capillary sprouts are formed which migrate towards the tumour, eventually penetrating it and permitting vascular growth to take place. This paper will present several mathematical models which deal with the various stages of growth and development of solid tumours.

Keywords—Avascular growth, Tumour angiogenesis, Vascular growth.

1. INTRODUCTION

In vivo cancer growth, and as a particular example in this paper, solid tumour growth, is a complicated phenomenon involving many interrelated processes and as such presents the mathematical modeller with a correspondingly complex set of problems to solve. Even deciding which simplifying assumptions to make is a nontrivial task. In this paper, we present a variety of mathematical models describing the various phases of solid tumour growth.

Solid tumours are known to progress through two distinct phases of growth—the avascular phase and the vascular phase [1,2]. The initial avascular growth phase can be studied in the laboratory by culturing cancer cells in the form of three-dimensional *multicell spheroids* [3–5]. It is well known that these spheroids, whether grown from established tumour cell lines or actual *in vivo* tumour specimens, possess growth kinetics which are very similar to *in vivo* tumours. Typically, these avascular nodules grow to a few millimetres in diameter. Cells towards the centre, being deprived of vital nutrients, die and give rise to a necrotic core. Proliferating cells can be found in the outer three to five cell layers. Lying between these two regions is a layer of quiescent cells, a proportion of which can be recruited into the outer layer of proliferating cells. Much experimental data has been gathered on the internal architecture of spheroids, and studies regarding the distribution of vital nutrients (e.g., oxygen) and metabolites within the spheroids have been carried out [3,6–22]. Indeed, recent research into the phenomenon of hypoxia [23] has meant that such studies are enjoying a new lease of life.

The transition from the dormant avascular state to the vascular state, wherein the tumour possesses the ability to invade surrounding tissue and metastasise to distant parts of the body,

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depends upon its ability to induce new blood vessels from the surrounding tissue to sprout towards and then gradually penetrate the tumour, thus providing it with an adequate blood supply and microcirculation. In order to accomplish this neovascularization, it is now a well-established fact that tumours secrete a number of diffusible chemical substances into the surrounding tissues and extracellular matrix. Much work has been carried out into the nature of such substances and their effect on endothelial cells since initial research began in the early 1970's with Folkman, culminating in the purification of several angiogenic factors, the cloning of their genes from libraries of complementary DNA (cDNA) and the determination of their amino acid sequences [24]. The extensive current literature on the subject is testimony to its importance in our understanding of the mechanisms by which solid tumours develop and grow (see, for example, the reviews of Folkman and Klagsbrun [24] and Paweletz and Knierim [25]).

The first events of angiogenesis are rearrangements and migration of EC situated in nearby vessels [26]. The main function of EC is in the lining of the different types of vessels such as venules and veins, arterioles and arteries, small lymphatic vessels and the thoracic duct. They form a single layer of flattened and extended cells, and the intercellular contacts are very tight. Large intercellular spaces are not visible and any easy penetration of the established layer of cells is impossible. Special processes must take place for the intra- and extravasation of different cellular elements of the blood or the lymphatic fluids and tumour cells. Even intravascular tumour cells have to induce the formation of gaps in the single layer of EC in order to leave the respective vessels. These cells are the principal characters in the drama of angiogenesis and are always centre stage [25].

In response to the angiogenic stimulus, EC in the neighbouring normal capillaries which do not possess a muscular sheath are activated to stimulate proteases and collagenases. The endothelial cells destroy their own basal lamina and start to migrate into the extracellular matrix. Small capillary sprouts are formed by accumulation of EC which are recruited from the parent vessel. The sprouts grow in length by migration of the endothelial cells [27]. The experimental results of Sholley *et al.* [27] demonstrated that EC are continually redistributed among sprouts, moving from one sprout to another. This permits the significant outgrowth of a network of sprouts even when cell proliferation is prevented [27]. At some distance from the tip of the sprout, the EC divide and proliferate to contribute to the number of migrating EC. The mitotic figures are only observed once the sprout is already growing out, and cell division is largely confined to a region just behind the sprout tip. Solid strands of EC are formed in the extracellular matrix. Lumina develop within these strands and mitosis continues.

Initially the sprouts arising from the parent vessel grow in a more or less parallel way to each other. They tend to incline toward each other at a definite distance from the origin when neighbouring sprouts run into one another and fuse to form loops or *anastomoses*. Both tip–tip and tip–branch anastomosis occur and the first signs of circulation can be recognised. From the primary loops, new buds and sprouts emerge and the process continues until the tumour is eventually penetrated.

Once vascularized, the tumours grow rapidly as exophytic masses. In certain types of cancer, e.g., carcinoma arising within an organ, this process typically consists of columns of cells projecting from the central mass of cells and extending into the surrounding tissue area. The local spread of these carcinoma often assume an irregular jagged shape. By the time a tumour has grown to a size whereby it can be detected by clinical means, there is a strong likelihood that it has already reached the vascular growth phase.

In this paper, we present a review of some recent mathematical models which attempt to describe the various stages and processes involved in solid tumour growth—the avascular growth phase, tumour angiogenesis (capillary sprout formation) and vascular growth—while also presenting some new developments. In the following section, we begin with a simple but general model for the growth of an avascular tumour nodule or multicell spheroid which focuses on the potential role played by one or more growth inhibitory factors and the effect of the spatial distribution

of this factor on cell mitosis. In Section 3, we present a model for the growth and development of capillary sprouts during tumour angiogenesis. The model focuses on the roles of endothelial cell proliferation and migration during angiogenesis. New results concerning 2-dimensional simulations are presented. Section 4 develops and extends the model of Section 2 by introducing a growth promoting factor into the mathematical model and assesses the potential implications this may have for vascular tumour growth and invasion using reaction-diffusion theory. In Section 5, a model for the growth and development of a solid tumour is presented which extends early work by Greenspan and focuses on physical variables such as internal (interstitial) pressure, cell–cell adhesion and nutrient supply. The relationship of this model with all the models of the previous sections is developed. In Section 6, the relative merits and demerits of each model are discussed and various concluding remarks are made.

2. AVASCULAR TUMOUR GROWTH: THE MULTICELL SPHEROID MODEL

The first mathematical model for the growth of a solid tumour which we present in this section focuses on the initial avascular stage of growth, wherein the tumour, or from an experimental point of view, the multicell spheroid, grows to a diffusion-limited size of a few millimetres in diameter.

To incorporate all of the known and observed experimental results into a mathematical model, even for an avascular tumour (multicell spheroid), would be a very formidable task indeed. Furthermore, from such a complicated model it would be difficult to focus on the roles of individual processes. Even deciding upon simplifying assumptions can be difficult. However, at minimum a realistic model of spheroid growth should include certain nonuniformities in the central processes of inhibition of mitosis, consumption of nutrients, cell proliferation; as well as the dependence of cell mitotic rate on growth inhibitor concentration, geometrical constraints and central necrosis.

Several papers [28–37] have focussed attention on the chemical inhibition of mitosis within multicell spheroids, the main assumption of the modelling being that a growth inhibitory factor (GIF) is produced within the spheroid in some prescribed spatially-dependent manner to reflect the observed cellular heterogeneity within spheroids.

The existence and properties of chemicals which inhibit mitosis are very well documented [38–44]. Indeed there is experimental evidence concerning the specific effects of extracts from necrotic areas in tumours. It has been demonstrated that these extracts possess the ability to reduce the proliferation of cultured cells [45] and also of monolayer tumour cells [14,15], and the same effects have been proposed to occur in tumours *in vivo* [46]. Initially we focus attention on the *diffusion* of a growth inhibitory factor within a multicell spheroid and its possible effect on cell mitosis and proliferation. To this end, instead of a nonlinear production term, we consider the effects of introducing a nonlinear, spatially-dependent diffusion coefficient. It is known that in spheroids normal structural associations (e.g., cell–cell junctions) are disrupted, and hence the possibility arises of normal intercellular signals being disrupted also. Nonlinear diffusion of the GIF within the spheroid is one possible and simple way to take account of this. The results of this approach are then compared and contrasted with those produced when a nonlinear production term is assumed.

The control of mitosis in tissues can be modelled as a schematic mechanism in which self-regulating growth is achieved as a result of negative feedback from the growing tissue [34]. In this approach mitosis is assumed to be controlled by a discontinuous switch-like mechanism, such that if the concentration of the (GIF) is less than some threshold level θ, say, in any region within the tissue, mitosis occurs in this region, whereas if the concentration is greater than θ, mitosis is completely inhibited. The differential equation describing the diffusion, production and degradation of the GIF within a spheroid can be written

$$\frac{\partial C}{\partial t} = D\nabla^2 C + f(C) + \lambda S(\mathbf{r}), \qquad \mathbf{r} \in \Omega, \tag{2.1}$$

where $C = C(\mathbf{r}, t)$ is the concentration of GIF within the spheroid occupying the region $\Omega \in \mathcal{R}^3$ and λ is the inhibitor production rate (molecules per unit volume per second).

A more detailed description of the background biology and the derivation of the above equation can be found in [28–34]. Previous papers make the assumptions that the GIF is produced throughout the tissue by the individual cells, modelled by a source function $S(\mathbf{r})$, diffuses with a *constant* diffusion coefficient D and is depleted everywhere at a prescribed rate $f(C)$. Adam [28–30] and Shymko and Glass [34] use the function $f(C) = -\gamma C$, where γ is the decay or loss constant. Various forms for the source function $S(\mathbf{r})$ have been used. In the original model of Shymko and Glass [34], the GIF is assumed to be produced at a constant rate throughout the tissue yielding the *uniform* source function

$$S(\mathbf{r}) = \begin{cases} 1, & \mathbf{r} \text{ inside the tissue,} \\ 0, & \mathbf{r} \text{ outside the tissue.} \end{cases} \tag{2.2}$$

In an attempt to model more accurately the heterogeneity of cells within tumours, the models of Adam [28–30] consider a *nonuniform* source function of the form

$$S(r) = \begin{cases} 1 - \dfrac{r}{R}, & 0 \le r \le R, \\ 0, & r > R, \end{cases} \tag{2.3}$$

where r is the distance from the origin. However, as shown by Britton and Chaplain [47], the nonuniform source term of Adam [28–30] is unrealistic from a biological point of view since $S'(0) \neq 0$, and they suggest a smooth source function of the form

$$S(r) = \begin{cases} 1 - \dfrac{r^2}{R^2}, & 0 \le r \le R, \\ 0, & r > R. \end{cases} \tag{2.4}$$

If the above system is studied in a spherical geometry, with radial symmetry, that is, a sphere of radius R, where the GIF concentration depends only on the radial distance from the centre of the sphere ($C \equiv C(r)$), then it may be used to model mitotic inhibition within multicell spheroids. As stated in the introduction, there is a large body of experimental data available for multicell spheroids which can be used to study tumour microregions, the results of which can be applied to *in vivo* tumours [4,16]. Small spheroidal nodules of tumour cells can be grown in the laboratory, the growth kinetics of which closely follow the first avascular stage of *in vivo* tumour growth such as carcinomas. These grow to a diffusion-limited steady-state size a few millimetres in diameter. Cells at the centre of the avascular nodule are starved of vital nutrients and die [48], forming a central necrotic core. A thin layer of live, proliferating cells three to five cell layers thick forms at the periphery of the nodule while sandwiched between these two regions is a layer of quiescent cells. A proportion of the quiescent cells is reproductively viable and may be recruited into the proliferating compartment. One feature of spheroid growth which has hitherto not been treated in detail by mathematical models is the importance of cell–cell contacts. It is known that spheroids are held together by various surface membrane microprojections, extracellular matrix and a variety of cell–cell junctions including spot desmosomes, tight junctions, junctional complexes and gap junctions. It is possible that the loss of coupling between cells may play a critical role in the uncontrolled proliferation in cancer [49]. In addition to this, whether through gap junctions or by other structures, intercellular permeability may not be constant between cells at different stages of spheroid growth [7,8,50]. These observations suggest that diffusion of chemicals between cells may not be constant [10]. In this paper therefore, we examine the effect of spatial nonuniformity not on the production term but on the diffusion coefficient, and to this

end the model we consider is

$$\frac{\partial C}{\partial t} = \nabla.\,(D(\mathbf{r})\nabla C) + f(C) + \lambda S(\mathbf{r}), \qquad \mathbf{r} \in \Omega, \tag{2.5}$$

$$\frac{\partial C}{\partial r} = 0, \qquad \mathbf{r} = \mathbf{0}, \tag{2.6}$$

$$D(\mathbf{r})\frac{\partial C}{\partial n} + PC = 0, \qquad \text{on } \partial\Omega, \quad P \geq 0, \tag{2.7}$$

where we assume (initially) that production of GIF is via the uniform source function given in (2.2); P is the permeability of the tissue surface, and the diffusion coefficient $D(\mathbf{r})$ in the above equation is now assumed to be spatially dependent, i.e., it varies throughout the interior of the spheroid, for which there is experimental evidence [51]. Thus the observed cellular heterogeneity within spheroids, the heterogeneous intercellular permeability, and the composition of the extracellular matrix is accounted for via nonlinear diffusion. The boundary conditions (2.6),(2.7) have been dealt with at length elsewhere [28–30,34] and we will not discuss their derivation here.

Considering the spherical geometry described in the introduction and assuming radial symmetry, the above system reduces to

$$\frac{\partial C}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 D(r)\frac{\partial C}{\partial r}\right) - \gamma C + \lambda, \qquad r \leq R, \tag{2.8}$$

$$\frac{\partial C}{\partial r} = 0, \qquad r = 0, \tag{2.9}$$

$$D(r)\frac{\partial C}{\partial r} + PC = 0, \qquad r = R. \tag{2.10}$$

In order to facilitate analysis of the above system, it is convenient to nondimensionalise, and hence, we choose appropriate reference variables. The obvious variables to choose [32] are, as the reference length R, the radius of the spheroid under consideration, as a reference GIF concentration, θ the critical threshold level of C, and as a reference time R^2/D. Thus we define new variables

$$\tilde{r} = \frac{r}{R}, \qquad \tilde{C} = \frac{C}{\theta}, \qquad \tilde{t} = \frac{Dt}{R^2}.$$

We also assume that we can write

$$D(r) = Dd(r), \tag{2.11}$$

where D is a constant. The system now becomes, upon dropping the tildes for notational convenience,

$$\frac{\partial C}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 d(r)\frac{\partial C}{\partial r}\right) - L^2 C + aL^2, \tag{2.12}$$

$$\frac{\partial C}{\partial r} = 0, \qquad r = 0, \tag{2.13}$$

$$d(r)\frac{\partial C}{\partial r} + \frac{L}{\eta}C = 0, \qquad r = 1, \tag{2.14}$$

where $\kappa^2 = \gamma/D$, $L = \kappa R$, $a = \lambda/\gamma\theta$ and $\eta = ((\gamma D)^{1/2})/P$. With this nondimensionalisation, we see that once the parameters for a particular spheroid are determined, the only undetermined parameter is the radius R. The solutions to the above equation can therefore be monitored for different sized spheroids. Thus we can analyse the system using different values for the spheroid radius R while holding constant the various observable parameters associated with the system, i.e., D, γ, P, and λ/θ [34]. It is our intention to discover if a concentration profile of GIF inside the spheroid can be found corresponding to the observed pattern of necrosis seen experimentally.

Since it is known that rapidly growing tumours enlarge at a rate of the order 0.4–0.7 mm/day [52], while diffusing chemical substances, with diffusion coefficients of order $10^{-6}\,\mathrm{cm}^2\,\mathrm{s}^{-1}$, reach steady-state on a time-scale of the order of a few minutes, then it is sufficient to consider the corresponding steady-state equation given by

$$0 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 d(r)\frac{\partial C}{\partial r}\right) - L^2 C + aL^2, \tag{2.15}$$

together with the boundary conditions (2.13),(2.14). In order to accomplish this, (2.12) was solved using the NAG routine D03PGF, with zero initial conditions $C(r,0) = 0$. The steady-state solutions evolved on a time scale of $O(1)$. Different forms for the function $d(r)$ were considered, which we assume is some monotonic function of r. Both monotonically increasing functions of $d(r)$ ($d(0) < d(1)$) and monotonically decreasing functions ($d(0) > d(1)$) were investigated. For our purposes, we chose $d(r) = 0.8 + 0.2r^2$ as an example of the former and $d(r) = 1.0 - 0.2r^2$ as an example of the latter (variability in diffusion constants between spheroids grown from different cell lines has been verified experimentally [10]). As stated above, once the observable parameters have been fixed, the concentration profile for different sized spheroids can be easily obtained from the steady-state solutions of (2.12) simply by varying one parameter, namely the spheroid radius R. Mathematical models for this stage of spheroid growth involving the dynamic evolution of the radius $R(t)$ can be found in, for example, [53–57]. In these models, since the initial growth of the spheroids is avascular, they grow to a *finite*, diffusion-limited size a few millimetres in diameter. Mathematically, as $t \to \infty$, $R(t) \to R^*$. Thus by varying (i.e., increasing) the value for R in (2.12) from an initial value $R(0)$, say, up to R^*, we can effectively follow the development of the GIF concentration profile within the spheroid as the spheroid grows. This seems intuitively more appropriate rather than letting $R \to \infty$ which does not happen in practice. We also note that a consequence of the nondimensionalisation is that in each case the critical threshold level for mitosis occurs at $C = 1$.

2.1. Results

The results presented in this section are compared with the experimental data of Folkman and Hochberg [58]. In order to compare and contrast our results with previous mathematical models, following Chaplain and Britton [32] and Shymko and Glass [34], we choose $D = 5 \times 10^{-7}\,\mathrm{cm}^2\,\mathrm{s}^{-1}$, $P = 10^{-4}\,\mathrm{cm\,s}^{-1}$ and $\gamma = 5 \times 10^{-5}\,\mathrm{s}^{-1}$. Correspondingly, $\kappa = 10\,\mathrm{cm}^{-1}$ and $\eta = 0.05$. We note that in accordance with our assumption of a spatially varying diffusion coefficient (cf. (2.11)) the value of D given above will hold only at $r = 1$, i.e., the tumour edge. The value of the diffusion coefficient in the interior of the spheroid will vary according to the prescribed function $d(r)$. In order to find a value for the parameter a, we follow the procedure of Chaplain and Britton [32] and use the experimental observation of Folkman and Hochberg [58] that when the spheroids are approximately 0.05 cm in radius, the volume of the active mitotic zone is 0.6 of the total volume of the tissue. A simple calculation then shows that the size of the inner radius of the necrotic core must be 0.037 cm. In terms of the nondimensionalized variables, we therefore require that $C = 1$ when $r = 0.74$, $L = 0.5$, $\eta = 0.05$. Equation (2.12) was solved numerically with various values of the parameter a and allowed to evolve to steady-state. Figure 2.1 shows the steady-state profiles achieved with a value of $a = 32.5$, i.e., $\lambda/\theta = 1.625 \times 10^{-3}\,\mathrm{s}^{-1}$ for (i) a monotonically increasing and (ii) a monotonically decreasing diffusion coefficient, respectively. It can be seen from Figure 2.1 that this value of a (approximately) satisfies the above experimental constraint. Exact agreement could not be easily obtained since the analytic solution of (2.12) is not known. Figure 2.2 shows the steady-state GIF profiles within a spheroid of size 4 mm in diameter, i.e., the diffusion limited size. Once again the two curves correspond to (i) a monotonically increasing and (ii) a monotonically decreasing diffusion coefficient, respectively. The horizontal line drawn at $C = 1$ in all figures is the threshold value for the GIF. Hence in any region of the spheroid where

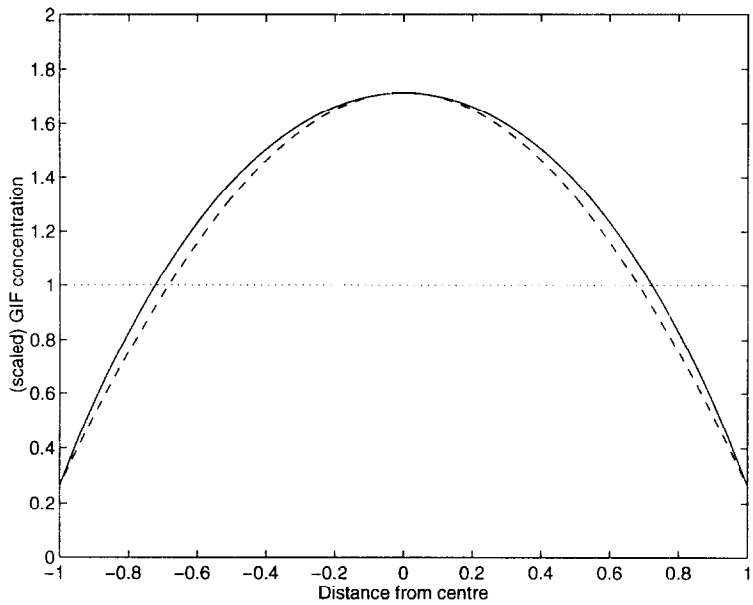

Figure 2.1. Plot of GIF concentration profile throughout multicell spheroid of size $R = 0.05\,\mathrm{cm}$; $D = 5 \times 10^{-7}\,\mathrm{cm}^2\,\mathrm{s}^{-1}$, $P = 10^{-4}\,\mathrm{cm}\,\mathrm{s}^{-1}$, $\gamma = 5 \times 10^{-5}\,\mathrm{s}^{-1}$; $a = 32.5$; $L = 0.5$; $\eta = 0.05$; (i) $d(r) = 0.8 + 0.2r^2$ (monotonically increasing), hatched line; (ii) $d(r) = 1.0 - 0.2r^2$ (monotonically decreasing), full line; constant source term.

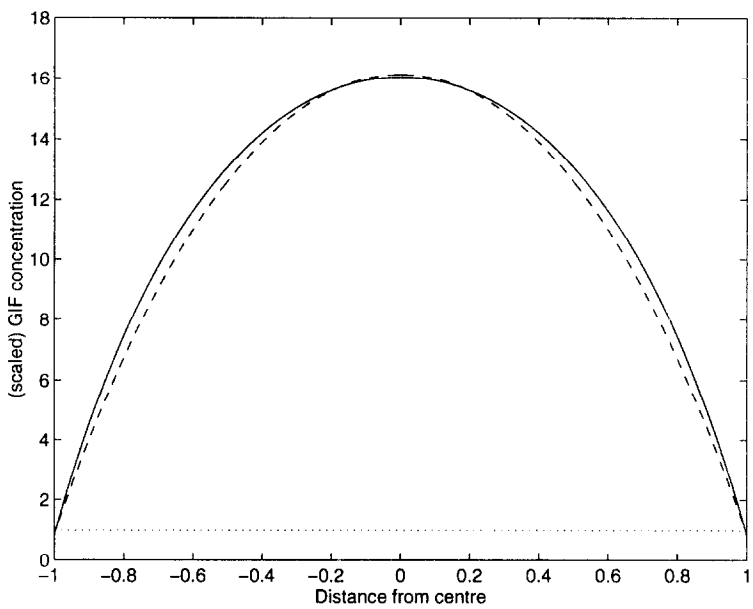

Figure 2.2. Plot of GIF concentration profile throughout multicell spheroid of ultimate size $R = 0.2\,\mathrm{cm}$; $D = 5 \times 10^{-7}\,\mathrm{cm}^2\,\mathrm{s}^{-1}$, $P = 10^{-4}\,\mathrm{cm}\,\mathrm{s}^{-1}$, $\gamma = 5 \times 10^{-5}\,\mathrm{s}^{-1}$; $a = 32.5$; $L = 2$; $\eta = 0.05$; (i) $d(r) = 0.8 + 0.2r^2$ (monotonically increasing), hatched line; (ii) $d(r) = 1.0 - 0.2r^2$ (monotonically decreasing), full line; constant source term.

the GIF concentration is greater than 1, then mitosis will be inhibited here. This enables regions within the spheroid where mitosis is taking place and where mitosis is inhibited (necrotic core) to be easily distinguished. As can be clearly seen from Figure 2.2, the final steady-state GIF profile achieved is in good agreement with the experimental data of Folkman and Hochberg [58] where at the stable, diffusion-limited size there was a narrow layer of one or two cells which was mitotically active surrounding the necrotic core. These results also parallel those of Chaplain and Britton [32].

We now also consider solving the above system with the constant source term being replaced by a spatially nonuniform source function of the type suggested by Chaplain and Britton [32,47], i.e.,

$$S(r) = \begin{cases} 1 - \dfrac{r^2}{R^2}, & 0 \le r \le R, \\ 0, & r > R. \end{cases} \tag{2.16}$$

After nondimensionalization, the model equation now becomes

$$\frac{\partial C}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 d(r)\frac{\partial C}{\partial r}\right) - L^2 C + aL^2(1 - r^2). \tag{2.17}$$

Figure 2.3 shows the steady-state GIF concentration profiles obtained with a value of $a = 65$ for (i) a monotonically increasing and (ii) a monotonically decreasing nonlinear diffusion term, respectively. Once again this value for a enables the experimental observation of Folkman and Hochberg [58] concerning the ratio of the volume of the active mitotic zone to the total tissue volume to be satisfied. Figure 2.4 shows the corresponding steady-state GIF concentration profiles for a spheroid of ultimate size 0.4 cm diameter. Again these show good agreement with the experimental observations.

We note for completeness that similar results to those of Britton and Chaplain [47] can be obtained upon straightforward application of the maximum principle to (2.12)–(2.14) and (2.17). Using these techniques, it is easily shown that, qualitatively, the final GIF concentration profile depends only on the qualitative form of the function $d(r)$ which can be chosen to model the interior heterogeneity of the spheroids.

3. TUMOUR ANGIOGENESIS—CAPILLARY SPROUT FORMATION AND GROWTH

In this section and throughout the rest of the paper wherever angiogenesis is mentioned, we choose to focus on the model system of Gimbrone *et al.* [59] and Muthukkaruppan *et al.* [60] wherein a fragment of tumour is implanted into the cornea of a test animal, and an angiogenic response is elicited from blood vessels situated in the nearby limbus region of the cornea. The mathematical model presented here is a development and extension of that of Chaplain and Stuart [61].

Tumour angiogenesis factor having concentration $c(\mathbf{x}, t)$ is secreted by the solid tumour and diffuses into the surrounding tissue. Upon reaching neighbouring endothelial cells situated in, for example, the limbal vessels, the TAF stimulates the release of enzymes by the endothelial cells which degrade their basement membrane. As described in the introduction, after degradation of the basement membrane has taken place, the initial response of the endothelial cells is to begin to migrate towards the source of angiogenic stimulus. Capillary sprouts are formed and cells subsequently begin to proliferate at a later stage. Once the capillary sprouts have formed, mitosis is largely confined to a region a short distance behind the sprout tips [25,27,62,63]. In [62], it was hypothesised that the reason for this proliferation was that these cells or vessels at the sprout tips were *acting as sinks* for the TAF. Balding and McElwain [64] also suggested that a sink term could be included in their model of capillary growth. Following Chaplain and Stuart [61,65], we thus incorporate a sink term for the TAF *in addition to* a natural decay term for the TAF. The conservation equation for the TAF concentration is thus given by

$$\begin{array}{c}\text{rate of increase}\\ \text{of TAF}\end{array} = \begin{array}{c}\text{diffusion}\\ \text{of TAF}\end{array} - \begin{array}{c}\text{loss due to}\\ \text{cells}\end{array} - \begin{array}{c}\text{decay of}\\ \text{chemical}\end{array}$$

which, under the assumption of linear Fickian diffusion, can be written mathematically as

$$\frac{\partial c}{\partial t} = D_c \nabla^2 c - f(c)g(n) - h(c), \tag{3.1}$$

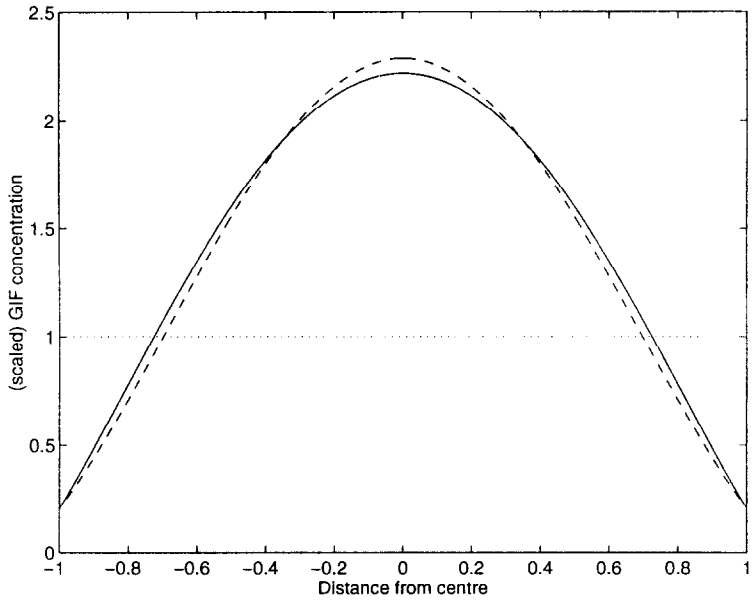

Figure 2.3. Plot of GIF concentration profile throughout multicell spheroid of size $R = 0.05\,\mathrm{cm}$; $D = 5 \times 10^{-7}\,\mathrm{cm^2\,s^{-1}}$, $P = 10^{-4}\mathrm{cm\,s^{-1}}$, $\gamma = 5 \times 10^{-5}\,\mathrm{s^{-1}}$; $a = 65$; $L = 0.5$; $\eta = 0.05$; (i) $d(r) = 0.8 + 0.2r^2$ (monotonically increasing), hatched line; (ii) $d(r) = 1.0 - 0.2r^2$ (monotonically decreasing), full line; nonlinear source term.

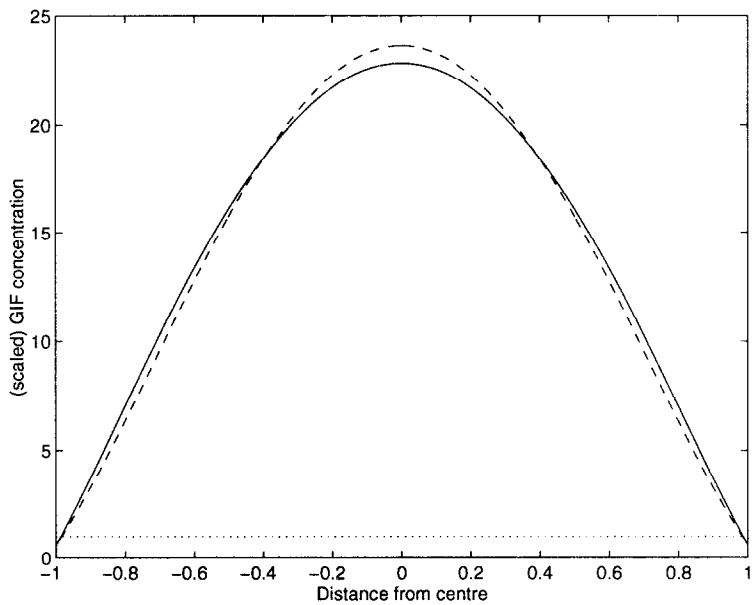

Figure 2.4. Plot of GIF concentration profile throughout multicell spheroid of ultimate size $R = 0.2\,\mathrm{cm}$; $D = 5 \times 10^{-7}\,\mathrm{cm^2\,s^{-1}}$, $P = 10^{-4}\,\mathrm{cm\,s^{-1}}$, $\gamma = 5 \times 10^{-5}\,\mathrm{s^{-1}}$; $a = 65$; $L = 2$; $\eta = 0.05$; (i) $d(r) = 0.8 + 0.2r^2$ (monotonically increasing), hatched line; (ii) $d(r) = 1.0 - 0.2r^2$ (monotonically decreasing), full line; nonlinear source term.

where D_c is the TAF diffusion coefficient. We assume that the local rate of uptake of TAF by the endothelial cells (modelled by the function $f(c)$) is governed by Michaelis-Menten kinetics (cf. [61,65]) and that it also depends on the cell density, hence the inclusion of the function $g(n)$, i.e., the greater the density of endothelial cells, the more TAF will be removed by the cells acting as sinks (cf. [62,65]). In general, the function $g(n)$ can therefore be chosen to be some strictly increasing function to account for this. For simplicity, the actual function used in the model

is given by $g(n) = n/n_0$, a simple linear function. More complicated choices are also possible, however, and do not affect the model qualitatively. We also assume that the decay of TAF with time is governed by first-order kinetics, a standard assumption, cf. [66]. This leads to the following equation for the TAF in the external tissue:

$$\frac{\partial c}{\partial t} = D_c \nabla^2 c - \frac{Qcn}{(K_m + c)\, n_0} - dc. \tag{3.2}$$

The initial condition is

$$c(\mathbf{x}, 0) = c_0(\mathbf{x}), \tag{3.3}$$

where $c_0(\mathbf{x})$ is a prescribed function chosen to describe qualitatively the profile of TAF in the external tissue when it reaches the limbal vessels, cf. [65]. The TAF is assumed to have a constant value c_b on the boundary of the tumour and to have decayed to zero at the limbus [65] giving the boundary conditions as

$$c(0, t) = c_b, \qquad c(L, t) = 0. \tag{3.4}$$

Endothelial cell population balance equation

The endothelial cells are the principal characters in the drama of angiogenesis and are always centre stage [25]. It is therefore highly desirable and logical to include in our model an equation modelling the endothelial cells. The cascade of events which goes to make up the complex process of angiogenesis is essentially driven by the endothelial cells. We will thus follow the route of the endothelial cells from their origin in their parent vessel (e.g., the limbus), their crossing of the extracellular matrix and other material in the surrounding host tissue, to their destination within the tumour.

The first events of angiogenesis are rearrangements and migration of endothelial cells rather than induction of cell division [25,26]. In response to the angiogenic stimulus, endothelial cells in the neighbouring normal capillaries which do not possess a muscular sheath are activated to stimulate proteases and collagenases. The endothelial cells destroy their own basal lamina and start to migrate into the extracellular matrix. Small capillary sprouts are formed by accumulation of endothelial cells which are recruited from the parent vessel. The sprouts grow in length by migration of the endothelial cells [27,67–69]. The experimental evidence of Sholley *et al.* [27] demonstrated that endothelial cells are continually redistributed among sprouts, moving from one sprout to another. This permits the significant outgrowth of a network of sprouts even when cell proliferation is prevented [27]. At some distance from the tip of the sprout, the endothelial cells divide and proliferate to contribute to the number of migrating endothelial cells. The mitotic figures are only observed once the sprout is already growing out and cell division is largely confined to a region just behind the sprout tip. Solid strands of endothelial cells are formed in the extracellular matrix. Lumina develop within these strands and mitosis continues.

Initially, the sprouts arising from the parent vessel grow in a more or less parallel way to each other. They tend to incline toward each other at a definite distance from the origin when neighbouring sprouts run into one another and fuse to form loops or *anastomoses*. Both tip–tip and tip–branch anastomosis occur and the first signs of circulation can be recognised. From the primary loops, new buds and sprouts emerge and the process continues until the tumour is eventually penetrated.

We now attempt to account for the above sequence of events using a population balance equation for the endothelial cells and to interpret the processes described above by analysing the endothelial cell density profile. The main events we model are the *migration* and the *proliferation* of the endothelial cells (the processes of anastomosis and budding will be accounted for implicitly in the model). We note that the migration and replication of endothelial cells are not linked together. Different types of stimuli are necessary for these two processes, and we take this

important fact into account in our model. We begin then with a general conservation equation for the endothelial cell density $n(\mathbf{x}, t)$ which is of the form [61]

$$\frac{\partial n}{\partial t} + \nabla.\mathbf{J} = F(n)G(c) - H(n), \tag{3.5}$$

where $\mathbf{J}$ is the cell flux, $F(n)$ and $H(n)$ are functions representing a normalised growth term and a loss term, respectively, for the endothelial cells. We assume that mitosis is governed by logistic type growth and that cell loss is a first order process [27,61]. Thus

$$F(n) = rn\left(1 - \frac{n}{n_0}\right), \tag{3.6}$$

$$H(n) = -k_p n, \tag{3.7}$$

where r is a positive constant related to the maximum mitotic rate and k_p is the proliferation rate constant which is taken to be the reciprocal of the endothelial cell doubling time [63,66]. We note that (3.6) contains a second order loss term while (3.7) is a first order loss term. Balding and McElwain [64] considered cell loss due to anastomosis (both tip–tip and tip–branch) as essentially a second order process, while Stokes and Lauffenburger [63] modelled endothelial cell loss due to budding as a first order loss term. Moreover, they assumed that the probability of budding was uniform in all sprouts for all positions and all times. Thus these two terms implicitly account for endothelial cell loss due to anastomosis and budding, respectively. Further we assume that the endothelial cell proliferation is controlled in some way by the TAF [25,26] and this is reflected by the inclusion of the function $G(c)$ which is assumed to be nondecreasing. As stated previously, the initial response of endothelial cells to the angiogenic stimulus is one of migration [25,26]. Proliferation is a crucial but secondary response. In order to account for this through the function $G(c)$, we assume that there is a threshold concentration level of TAF below which proliferation does not occur. Thus in the present model, we chose $G(c)$ to be of the form

$$G(c) = \begin{cases} 0, & c \leq c^*, \\ \dfrac{c - c^*}{c_b}, & c^* < c, \end{cases} \tag{3.8}$$

where $c^* \leq c_b$. We note that a similar approach was used by Sherratt and Murray [66] in modelling the chemical control of epithelial cells during wound healing.

There is substantial evidence that the response of the endothelial cells to the presence of the TAF is a chemotactic one [62,63], and following Chaplain and Stuart [61], we assume that the flux $\mathbf{J}$ of endothelial cells consists of two parts, one representing random motion and the other chemotactic motion of the cells. Thus

$$\mathbf{J} = \mathbf{J}_{\text{diffusion}} + \mathbf{J}_{\text{chemotaxis}}. \tag{3.9}$$

Once again we assume linear diffusion so that

$$\mathbf{J}_{\text{diffusion}} = -D_n \nabla n, \tag{3.10}$$

where D_n is the diffusion coefficient of the endothelial cells and

$$\mathbf{J}_{\text{chemotaxis}} = n\chi(c)\nabla c, \tag{3.11}$$

the well-known form for the chemotactic flux. Various functional forms have been proposed for $\chi(c)$ including a logarithmic law

$$\chi(c) = \frac{\chi_0}{c},$$

a receptor kinetic law

$$\chi(c) = \frac{\chi_0 k}{(k+c)^2},$$

and a constant law

$$\chi(c) = \chi_0 \ \text{(a constant)}.$$

For mathematical simplicity, throughout this paper we adopt the latter, i.e., we take $\chi(c) = \chi_0$, a constant. We note however that the present model could easily be adapted to incorporate either the logarithmic law or the receptor kinetic law. The cell conservation equation can be written

$$\begin{matrix}\text{rate of increase}\\ \text{of cell density}\end{matrix} = \begin{matrix}\text{cell}\\ \text{migration}\end{matrix} + \begin{matrix}\text{mitotic}\\ \text{generation}\end{matrix} - \begin{matrix}\text{cell}\\ \text{loss}\end{matrix}.$$

With the above assumptions, we thus have the following population diffusion-chemotaxis equation for the endothelial cells:

$$\frac{\partial n}{\partial t} = D_n \nabla^2 n - \chi_0 \nabla.(n \nabla c) + rn\left(1 - \frac{n}{n_0}\right) G(c) - k_p n, \tag{3.12}$$

where $G(c)$ is given by (3.8). We assume that initially the endothelial cell density at the limbus is a constant n_0 and zero elsewhere, giving initial condition

$$n(\mathbf{x}, 0) = \begin{cases} n_0, & x = L, \\ 0, & x < L. \end{cases} \tag{3.13}$$

We assume that throughout the subsequent motion, the cell density remains constant at the limbus and hence the boundary condition here becomes

$$n(L, t) = n_0. \tag{3.14}$$

As stated previously, the main aim of the model is to monitor the progress of the endothelial cells (in particular those at the sprout tips) as they cross the ECM and eventually reach the tumour. Once they reach the tumour and penetrate it, interactions with the tumour cells become important [25] and the assumptions of the present model no longer hold. The modelling of this stage of the process is considered by Liotta *et al.* [70] and Chaplain and Sleeman [71]. Thus within the assumptions and limitations of the present model, we consider the following boundary condition at $\mathbf{x} = \mathbf{0}$:

$$n = 0 \qquad \text{at } \mathbf{x} = \mathbf{0}. \tag{3.15}$$

Under the assumptions of the model, this boundary condition will remain valid up to the time when the endothelial cells at the sprout tips first reach the tumour.

Following Chaplain and Stuart [61,65], we normalize the equations using the following reference variables:

- reference TAF concentration: c_b, the value of the TAF concentration at the tumour boundary,
- reference cell density: n_0, the value of the endothelial cell density at the limbus,
- reference length: L, the distance from the tumour boundary to the limbal vessels,
- reference time unit: $\tau = L^2/D$.

We thus define new variables:

$$\tilde{c} = \frac{c}{c_b}, \qquad \tilde{n} = \frac{n}{n_0}, \qquad \tilde{x} = \frac{x}{L}, \qquad \tilde{t} = \frac{t}{\tau}.$$

Dropping the tildes and specialising to a one-dimensional geometry, the equations now become

$$\frac{\partial c}{\partial t} = \frac{\partial^2 c}{\partial x^2} - \frac{\alpha n c}{\gamma + c} - \lambda c, \tag{3.16}$$

$$\frac{\partial n}{\partial t} = D\frac{\partial^2 n}{\partial x^2} - \kappa\frac{\partial}{\partial x}\left(n\frac{\partial c}{\partial x}\right) + \mu n(1-n)G(c) - \beta n, \tag{3.17}$$

where

$$G(c) = \begin{cases} 0, & c \le c^*, \\ c - c^*, & c^* < c, \end{cases} \tag{3.18}$$

$$\begin{aligned} \alpha &= \frac{L^2 Q}{D_c c_b}, & \gamma &= \frac{K_m}{c_b}, & \lambda &= \frac{L^2 d}{D_c}, \\ D &= \frac{D_n}{D_c}, & \kappa &= \frac{c_b \chi_0}{D_c}, & \mu &= \frac{L^2 r}{D_c}, \\ \beta &= \frac{L^2 k_p}{D_c}. \end{aligned} \tag{3.19}$$

The initial and boundary conditions become, respectively,

$$c(x,0) = c_0(x), \tag{3.20}$$

$$n(x,0) = \begin{cases} 1, & x = 1, \\ 0, & x < 1, \end{cases} \tag{3.21}$$

$$c(0,t) = 1, \qquad c(1,t) = 0, \tag{3.22}$$

$$n(1,t) = 1, \tag{3.23}$$

$$n = 0 \qquad \text{at } x = 0. \tag{3.24}$$

As we have mentioned previously, angiogenic factors must be able to provoke three main activities of the endothelial cells, namely

(1) production and secretion of enzymes capable of digesting extracellular matrix,
(2) the initiation of *migration*, and
(3) cell *proliferation*.

It is clear from experimental evidence [27,72,73] that *different* types of stimuli are necessary for (2) and (3), and we note that this is accounted for in the model, i.e., migration is governed by chemotaxis while proliferation is governed by the function $G(c)$ which essentially only depends on the TAF concentration. As stated previously, the processes of anastomosis and budding are also (implicitly) accounted for through the second and first order loss terms, respectively, in (3.17). Moreover, the form of the function $G(c)$ ensures that the second order loss term implicitly modelling anastomosis will only take effect after the sprouts have reached a certain distance into the extracellular matrix, which is precisely what is observed experimentally [25,26].

In this section, we present the results from the numerical simulation of (3.16)–(3.23) with appropriate boundary and initial conditions. As far as possible, parameter values are chosen to correspond to available experimental data.

For a reference length we choose $L = 2\,\text{mm}$, an average distance between a tumour implant and the limbal vessels, and the value for τ is taken to be 14 days, an average time for neovascularization to occur [63,64]. According to the nondimensionalisation in (3.19), this gives a value for D_c of $3.3\times 10^{-8}\,\text{cm}^2\,\text{s}^{-1}$. Sherratt and Murray [66] found values of $3.1\times 10^{-7}\,\text{cm}^2\,\text{s}^{-1}$ and $5.9\times 10^{-6}\,\text{cm}^2\,\text{s}^{-1}$ as estimates for diffusion coefficients of chemicals. Correspondingly, they estimated the diffusion coefficients D_n of the epithelial cells under consideration in their model as $3.5\times 10^{-10}\,\text{cm}^2\,\text{s}^{-1}$

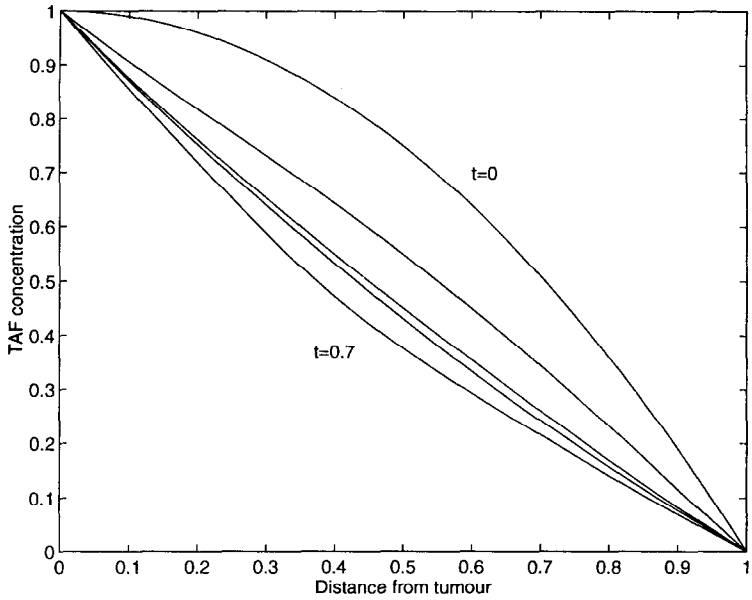

Figure 3.1. Profile of the TAF concentration in the external host tissue at times $t = 0, 0.1, 0.3, 0.5, 0.7$ showing changing gradient profile. Parameter values: $\alpha = 10$, $\gamma = 1$, $\lambda = 1$, $D = 0.001$, $\kappa = 0.75$, $\mu = 100$, $\beta = 4$, $c^* = 0.2$.

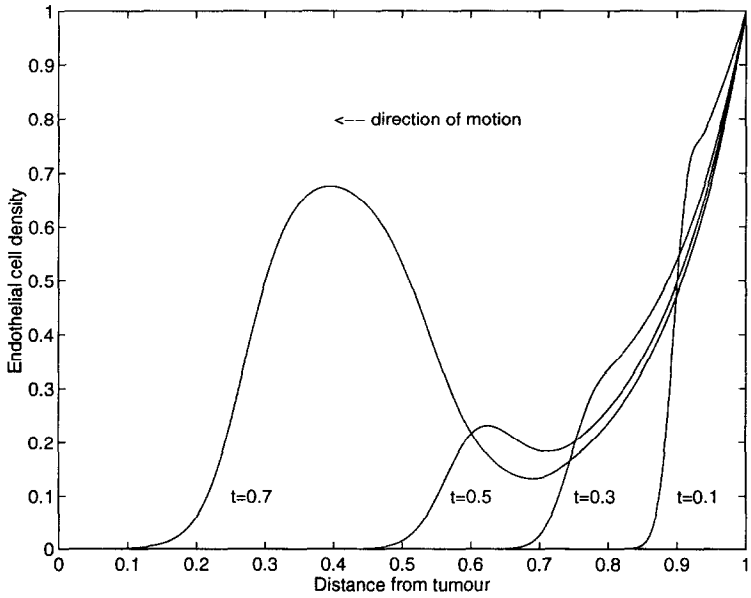

Figure 3.2. Profile of the endothelial cell density in the external host tissue at times $t = 0, 0.1, 0.3, 0.5, 0.7$. Shortly after $t = 0.7$, the endothelial cells reach the tumour and the assumptions of the model no longer hold. Parameter values per Figure 3.1.

and $6.9 \times 10^{-11}\,\mathrm{cm}^2\,\mathrm{s}^{-1}$. Using these values, the ratio D_n/D_c thus varies between 1.1×10^{-3} and 1.2×10^{-5}. In accordance with this range, we choose D to be 10^{-3}.

Stokes *et al.* [74] have measured the value for the chemotaxis coefficient χ_0 of endothelial cells migrating in a medium containing unpurified acidic fibroblast growth factor (aFGF) as $2600\,\mathrm{cm}^2\,\mathrm{s}^{-1}\,\mathrm{M}^{-1}$. As above, the value for D_c is estimated at $3.3 \times 10^{-8}\,\mathrm{cm}^2\,\mathrm{s}^{-1}$. In the numerical simulations carried out, the (nonzero) values for κ varied between 0.3 and 1.0, which from (3.19) gives a value of c_b of approximately $10^{-11}\,\mathrm{M}$ which is not unreasonable.

From equation (3.12), we assume that the parameter r is a positive constant related to the maximum mitotic rate. Using data based on epidermal wound healing [75], Sherratt and Mur-

ray [66] estimated that the maximum value $r_{\max}$ for this parameter was 10 times the proliferation rate constant. Based on *in vitro* experiments on endothelial cell proliferation [76], Stokes and Lauffenburger [63] estimated the proliferation rate constant k_p of endothelial cells to be $0.056\,\mathrm{hr}^{-1}$ under the assumption that all cells proliferate. However, cell mitosis in the sprouts is mainly confined to a region close to the tips. To compensate for this, in most of their simulations Stokes and Lauffenburger reduced the value of k_p to $0.02\,\mathrm{hr}^{-1}$ and assumed that all cells in a sprout may proliferate. Thus in the following numerical simulations, we assume a range for $r_{\max}$ of $0.2\,\mathrm{hr}^{-1}$–$0.56\,\mathrm{hr}^{-1}$. Under this assumption, this gives a range of values for the cell proliferation parameter μ of approximately 70–190.

The parameter k_p is the reciprocal of the endothelial cell doubling time (cf. [66]). As seen above, Sherratt and Murray [66] estimated this to lie within the range $0.02\,\mathrm{hr}^{-1}$–$0.056\,\mathrm{hr}^{-1}$. Also, from experimental data on epidermal cells [72], Sherratt and Murray [66] estimated this to be $0.01\,\mathrm{hr}^{-1}$. This gives a range of values for the cell loss parameter β of approximately 3–18.

The initial condition (3.20) was taken to be $c(x,0) = c_0(x) = 1 - x^2$ which is of the correct qualitative shape for the TAF profile in the external tissue, i.e., a constant value of 1 at the tumour edge decaying away to 0 at the limbus.

Figures 3.1, 3.2 show the profiles of the TAF concentration and the endothelial cell density, respectively, in the external host tissue at various different times. As can be seen from Figure 3.2, shortly after $t = 0.7$, the model loses its validity since the endothelial cells reach the tumour and interactions between tumour cells and endothelial cells become important. The time taken for the endothelial cells to first reach the tumour corresponds to a real time of approximately 11 days which is within the experimentally observed timescale, cf. [64]. By varying the parameters μ and κ, the time taken for the endothelial cells (and hence the capillary sprouts) to reach the tumour can also be varied.

Figure 3.3 shows the endothelial cell density profile when the chemotactic response parameter $\kappa = 0$, and as can clearly be seen, there is very little cell response, indicating that chemotaxis, and therefore cell migration, plays a major initiating role in the angiogenic process.

Figure 3.4 shows the endothelial cell density profile when the cell proliferation parameter $\mu = 0$, and once again the effect of this is significant. The figure shows that cell outgrowth, and hence sprout outgrowth, has virtually ceased after $t = 0.4$ which corresponds to a real time of 3.5 days. This is in very good agreement both with experimental evidence [27] and with the model of Stokes and Lauffenburger [63]. The previous two figures demonstrate that neither cell migration (via chemotaxis) nor cell proliferation alone is sufficient for a completion of angiogenesis which is in agreement with experimental observations. In order for a completion of angiogenesis, both events must be included in the model which is what is observed experimentally.

Comparing Figure 3.2 with Figure 3.4 demonstrates a good qualitative agreement with the available experimental evidence. Both simulations give the same profile of cell density for times $0 \le t \le 0.4$. This shows that the initial response of the endothelial cells is essentially one of migration with proliferation of the cells occurring at a later time. The form of the function $G(c)$ ensures that there is always a region within the sprouts where there is zero cell proliferation and also that once cells have started to proliferate the proliferation is mainly confined to a region a short distance behind the sprout tips. Thus the model distinguishes, as far as is possible within its limitations, between proliferating cells near the sprout tip and nonproliferating cells within the rest of the sprout. Also once the cells have started to proliferate, the endothelial cell density is greatest (locally) a short distance behind the sprout tips which is what is observed experimentally [25,27,62]. Similar results were obtained for different functions $G(c)$ which were of the same qualitative form as (3.8).

For all numerical simulations carried out, we took $c^* = 0.2$. Qualitatively similar results were obtained for various other values of c^* between 0.1 and 0.4. We note that other numerical simulations were carried out with $D = 0$ and $\alpha = 0$. In each case, the profiles of TAF concentration and endothelial cell density remained almost the same as when these parameters were nonzero. In the

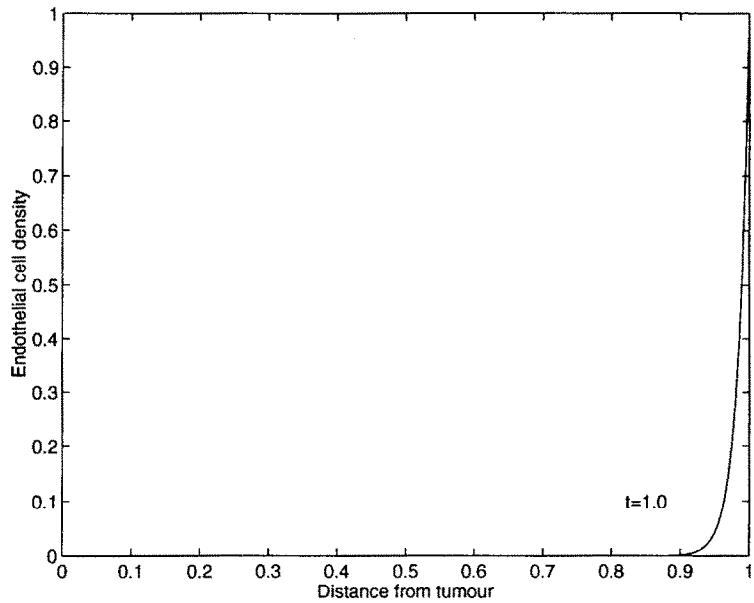

Figure 3.3. Profile of the endothelial cell density in the external host tissue at $t = 1.0$ when the chemotaxis parameter κ is set to zero. The figure shows very little cell, and hence sprout, growth. This illustrates that cell migration is vital to the complete process of angiogenesis. Parameter values: $\alpha = 10$, $\gamma = 1$, $\lambda = 1$, $D = 0.001$, $\kappa = 0.0$, $\mu = 100$, $\beta = 4$, $c^* = 0.2$.

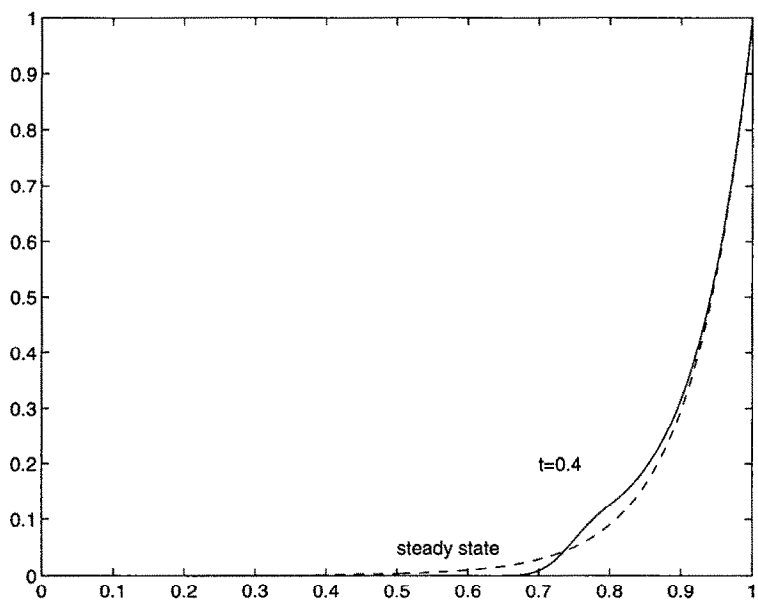

Figure 3.4. Profile of the endothelial cell density in the external host tissue at time $t = 0.4$ (full line) and at steady state (hatched line) when the cell proliferation parameter μ is set to zero. All outgrowth has stopped by steady state. The figure clearly shows that at $t = 0.4$ sprout outgrowth has virtually ceased, which is in agreement with the experimental evidence. This illustrates that cell proliferation is essential for the completion of angiogenesis. Parameter values: $\alpha = 10$, $\gamma = 1$, $\lambda = 1$, $D = 0.001$, $\kappa = 0.6$, $\mu = 0.0$, $\beta = 6$, $c^* = 0.2$.

former case, this would seem to confirm the results of the models of Stokes and Lauffenburger [63] and Balding and McElwain [64], where diffusion was also seen to have a negligible effect on the results.

3.1. Model Improvements and Extension

Endothelial cell population balance equation

The main events captured in the model of the previous section are the *migration* and the *proliferation* of the EC, with the important processes of *anastomosis* and secondary *budding/sprouting* being accounted for implicitly. We now extend the model by incorporating these two processes explicitly via the function $H(n)$. This function must now incorporate a (spatial) term which models cell loss due to secondary sprouting of tips from an existing capillary and at the same time model the effect of anastomosis (tip–tip or tip–branch) on a capillary. It is assumed that the net effect of anastomosis will be to increase the cell density at that particular location of the capillary where fusion takes place. Thus

$$H(n) = -k_p n - s(\mathbf{x})n, \tag{3.25}$$

where the function $s(\mathbf{x})$ will be taken to be some appropriate periodic function of the spatial variable $\mathbf{x}$ (e.g., $\cos(a\pi x)$), whose *average* value over a period is zero. Such a function will thus represent the effect of anastomosis for half its period (net increase in cell density due to cell gain) and the effect of secondary budding for the other half period (net reduction in cell density due to cell loss). This means that we assume secondary budding and anastomosis occur in a consecutive, periodic manner and is a reasonable approximation to what are essentially two random processes.

We also assume that the diffusion of the EC is density dependent which also generalises the model of Chaplain and Stuart [61]. The flux $\mathbf{J}$ is now written

$$\mathbf{J} = -D_n \left(\frac{n}{n_0}\right)^{\sigma} \nabla n + n\chi_0 \nabla c. \tag{3.26}$$

Simple Fickian diffusion is recovered on taking $\sigma = 0$. With the above assumptions, we thus have the following population diffusion-chemotaxis equation for the EC:

$$\frac{\partial n}{\partial t} = D_n \nabla.\left(\left(\frac{n}{n_0}\right)^{\sigma} \nabla n\right) - \chi_0 \nabla.(n\nabla c) + rn\left(1 - \frac{n}{n_0}\right) G(c) - k_p n - s(\mathbf{x})n. \tag{3.27}$$

Suitable initial and boundary conditions for the EC are prescribed depending upon the precise domain in which the above equations are solved. After a suitable normalization, the above equations are solved numerically and the results presented in the following section where the exact form of the boundary and initial conditions will also be given.

Finally, we also note that the numerical simulations of the previous section indicate that the concentration profile for c rapidly reaches a steady state profile. This is not entirely unexpected since the TAF diffuses much faster than the endothelial cells. Thus a reasonable approximation to the TAF concentration profile may be obtained by solving

$$\nabla^2 c - \lambda c = 0, \tag{3.28}$$

subject to $c = 1$ at tumour boundary, $c = 0$ at all other boundaries. This simplification means that now our model is reduced to a single partial differential equation

$$\frac{\partial n}{\partial t} = D_n \nabla.\left(n^{\sigma} \nabla n\right) - \kappa \nabla.(n\nabla c) + \mu n\left(1 - n\right) G(c(\mathbf{x})) - \beta n - s(\mathbf{x})n, \tag{3.29}$$

with appropriate initial and boundary conditions for n. This equation is solved numerically and the results are presented in the following section.

3.2. Numerical Simulations

We first solve our equation in a one-dimensional domain, normalized to be [0,1]. We assume the tumour implant is situated at $x = 0$ and the EC at $x = 1$. The solution to (3.28) (in one space dimension) is given by

$$c(x) = \frac{\sinh\left[(1-x)\sqrt{\lambda}\right]}{\sinh\sqrt{\lambda}}.$$

It can be shown that a good approximation to the TAF concentration profile with $\lambda = 1$ can be obtained on taking $c = 1 - x$. The equation to be solved is thus

$$\frac{\partial n}{\partial t} = D\frac{\partial}{\partial x}\left(n^\sigma \frac{\partial n}{\partial x}\right) + \kappa\frac{\partial n}{\partial x} + \mu(1-x)n(1-n) - \beta n - s(x)n, \tag{3.30}$$

with initial conditions

$$n(x,0) = \begin{cases} 1, & x = 1, \\ 0, & \text{elsewhere} \end{cases} \tag{3.31}$$

and boundary conditions $n(1,t) = 1$, $n(0,t) = 0$.

Figure 3.5 shows the endothelial cell density profile in the external tissue for two different functions $s(x)$. As can clearly be seen in each case, the results capture qualitatively both the effects of anastomosis (local maxima) and the formation of side branches (local minima) as the capillary sprout migrates towards the tumour. For profile (a) (hatched line – – –), we chose $s(x) = 5\cos(10\pi x)$ representing a regular spatial occurrence of both anastomosis and side-branching. However in profile (b) (solid line —), $s(x) = 5\cos(2.25\pi/x)$ which represents more faithfully the fact that as the capillary network progresses, anastomosis and side-branching occur more frequently. This captures more faithfully the so-called brush-border effect reported by Muthukkaruppan *et al.* [60].

Experiments are often carried out in very thin regions ($< 50\mu$m) of the cornea or ear chamber of test animals [59,60,63] and so a two-dimensional domain is a good approximation of the sprout growth. Using normalized variables, we work on the unit square $[0,1]\times[0,1]$ and assume that the tumour implant occupies the region $(x-0.5)^2 + y^2 \le 0.1$. We assume that the TAF diffuses in a radially symmetric manner, and it can easily be seen that the corresponding solution to (3.28) (in polar coordinates with radial symmetry) is given in terms of Bessel functions, but once again we find a very good approximation is given by

$$c(x,y) = c(r) = \begin{cases} 1, & r \le 1, \\ \dfrac{(1-r)^2}{0.81}, & r > 1, \end{cases} \tag{3.32}$$

where $r^2 = (x-0.5)^2 + y^2$. This profile is illustrated in Figure 3.6.

The single equation we now have to solve is given by

$$\frac{\partial n}{\partial t} = D\frac{\partial}{\partial x}\left(n^\sigma\frac{\partial n}{\partial x}\right) + D\frac{\partial}{\partial y}\left(n^\sigma\frac{\partial n}{\partial y}\right) - \kappa\frac{\partial}{\partial x}\left(n\frac{\partial c}{\partial x}\right) - \kappa\frac{\partial}{\partial y}\left(n\frac{\partial c}{\partial y}\right) + \mu c(x,y)n(1-n) - \beta n - s(x,y)n. \tag{3.33}$$

We assume that the initial phase of capillary sprout growth has taken place, i.e., the cells have secreted enzymes which degrade their basement membrane permitting the formation of small buds [25,26]. Indeed, Muthukkaruppan *et al.* [60] do not consider tumour-induced angiogenesis to be initiated until 2 days after implant and only designate a stage I of angiogenesis after 4 days recognising that there may well be some limbal response to the actual preparation of the implant site. Paweletz and Knierim [25] report two different initial mechanisms from which angiogenesis develops:

- Loops of vessels already present in the parental microcirculation are elongated towards the angiogenic stimulus (TAF) by inercalating EC. They always penetrate the tumour implant

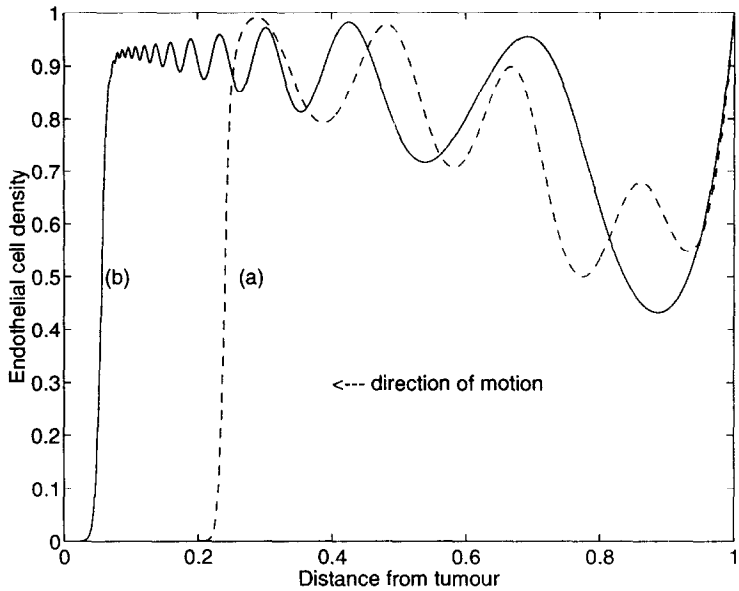

Figure 3.5. Profile of the endothelial cell density in the host tissue at a time corresponding to 12 days. Profile (a) (hatched line – – –) corresponds to regular anastomosis/side-branching with $s(x) = 5\cos(10\pi x)$, profile (b) (full line —) corresponds to more frequent anastomosis/side-branching with $s(x) = 5\cos(2.25\pi/x)$. Parameter values: $D = 0.001$, $\sigma = 4$, $\kappa = 0.65$, $\mu = 75$, $\beta = 5$.

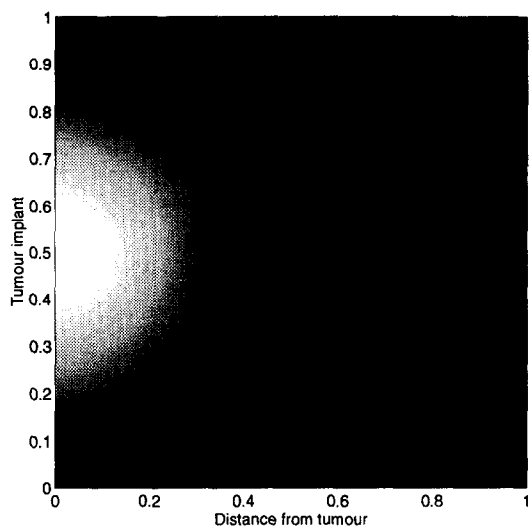

Figure 3.6. Profile of the steady state TAF concentration profile given by (3.32) used in the 2-dimensional simulations. White = high concentration, black = low.

as loops. Blood flow is possible within these elongating loops, but is realised only after completion of neovascularization. This kind of growth of new vessels apparently occurs after mild angiogenic stimulation and slow development of vascularization.

- During strong angiogenic induction and rapid growth, mostly sprouts which have to fuse at their tips to develop loops are formed.

We present initial conditions for each of the above scenarios. In the first case, we assume a loop (or loops) have already formed from pre-existing vessels (e.g., in the limbus) and assume that the endothelial cell density within this loop is initially a constant $n_0 = 1$ and zero elsewhere, giving initial conditions

$$n(\mathbf{x}, 0) = \begin{cases} n_0 = 1, & \mathbf{x} \text{ within loop}, \\ 0, & \mathbf{x} \text{ elsewhere}. \end{cases} \tag{3.34}$$

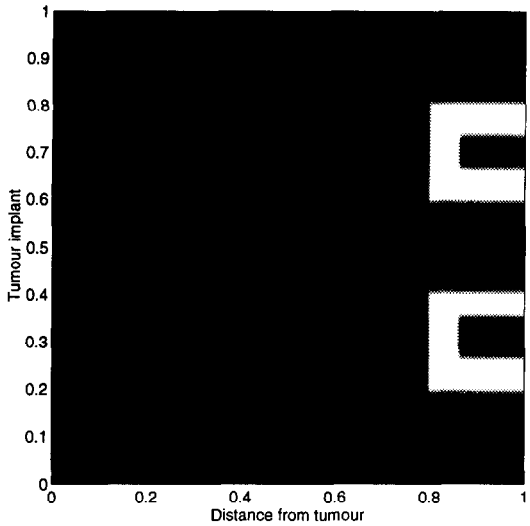

Figure 3.7. Profile of the initial conditions used for the endothelial cell density simulating the formation of two loops. White = high density, black = low.

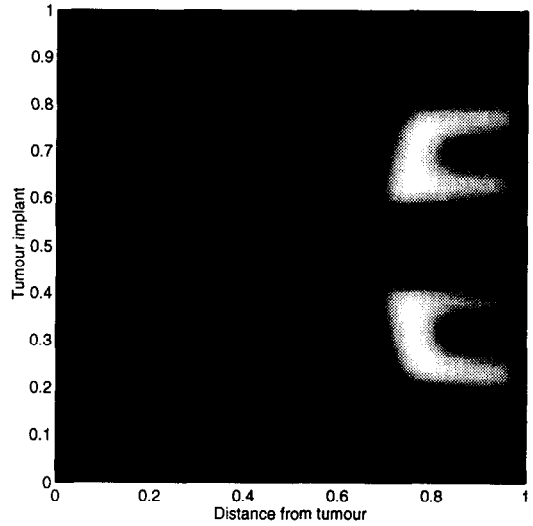

Figure 3.8. Profile of the endothelial cell density at time $t = 0.3$ showing evolution of the two loops. Parameter values: $D = 0.001$, $\sigma = 1$, $\kappa = 0.65$, $\mu = 75$, $\beta = 0.5$.

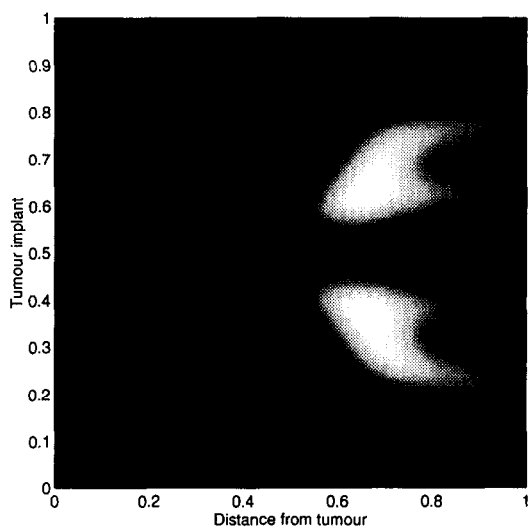

Figure 3.9. Profile of the endothelial cell density at time $t = 0.5$ showing further evolution of the two loops. Parameter values: $D = 0.001$, $\sigma = 1$, $\kappa = 0.65$, $\mu = 75$, $\beta = 0.5$.

In the second case, we assume that there are several capillary buds which have formed, and we focus attention exclusively upon those endothelial cells which are close to the sprout tips since only these cells proliferate. Here the endothelial cell density is initially a constant $n_0 = 1$ and zero elsewhere, giving initial conditions

$$n(\mathbf{x}, 0) = \begin{cases} n_0 = 1, & \mathbf{x} \text{ near sprout tip}, \\ 0, & \mathbf{x} \text{ elsewhere}. \end{cases} \tag{3.35}$$

Since we are attempting to focus attention on the EC near to the sprout tips (since these are the cells which proliferate) and also since all the EC are confined to within the capillary sprouts, we take the EC density to be zero ($n = 0$) on all boundaries.

Figures 3.7–3.9 show the evolution of two loops at times $t = 0, 0.3, 0.5$ as they migrate towards the tumour implant with initial conditions per (3.34). In this case, we chose $s(x, y) = 0$ since anastomosis and side-branching do not appear as frequently in this situation [25]. As can be seen

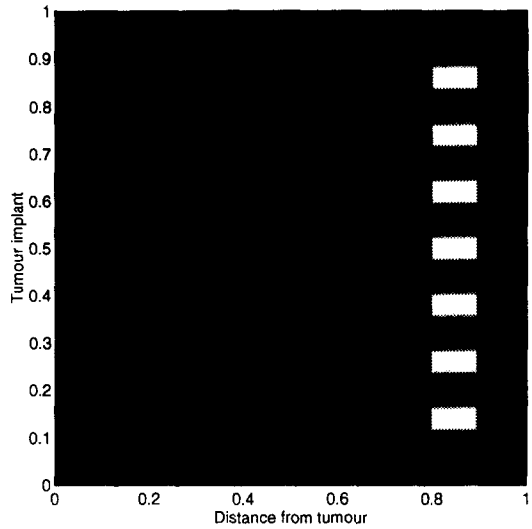

Figure 3.10. Profile of the initial conditions used for the endothelial cell density simulating the formation of several capillary sprouts. White = high density, black = low.

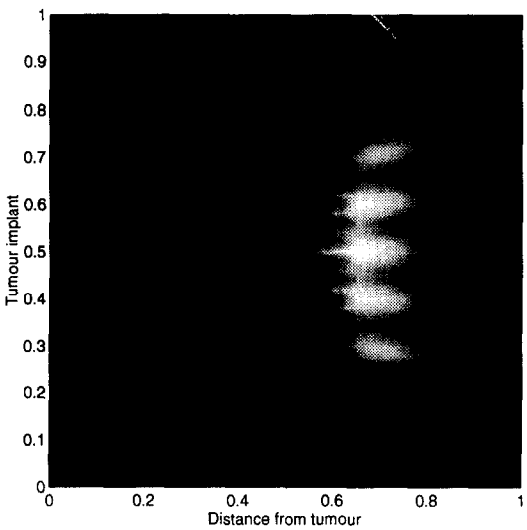

Figure 3.11. Profile of the endothelial cell density after the capillary network has been established and anastomosis has taken place corresponding to a real time of 5–6 days after initial sprouts have formed per Figure 3.9. Parameter values: $D = 0.001$, $\sigma = 1$, $\kappa = 0.65$, $\mu = 75$, $\beta = 0.5$.

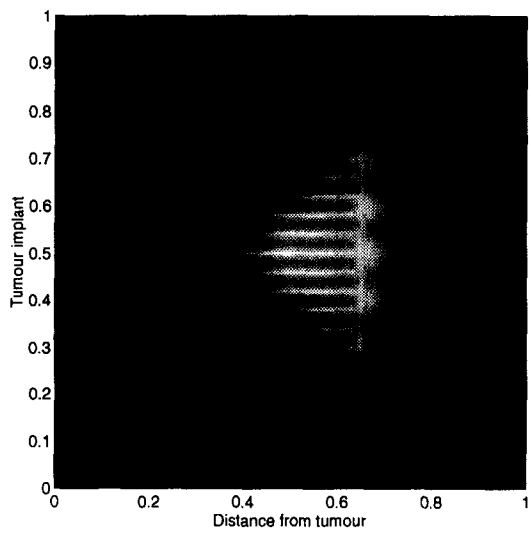

Figure 3.12. Profile of the endothelial cell density after the capillary network has been established and anastomosis has taken place corresponding to a real time of 7–8 days after initial sprouts have formed. Brush-border effect is clearly visible at the vascular front. Parameter values: $D = 0.001$, $\sigma = 1$, $\kappa = 0.65$, $\mu = 75$, $\beta = 0.5$.

from the figures, the loops continue to develop and progress as loops. Further figures at later times show that they penetrate the tumour as loops which is what is observed experimentally [25].

Figures 3.10–3.12 show the EC density profile within the capillary sprouts as they migrate towards the tumour with initial conditions per (3.35). In this case, we chose $s(x, y) = 10\cos^6(25\pi x)$, $y < 0.65$; $s(x, y) = 0$, $y \geq 0.65$, representing explicitly the effect of the brush-border once the initial sprouts/loops have all anastomosed, cf. [60].

4. VASCULAR TUMOUR GROWTH

While the process of angiogenesis is taking place, the avascular tumour, although dormant (or quasi-dormant) with regard to its growth, is still in a dynamic state. In this section, we consider the possibility of the development of a heterogeneous cellular prepattern which takes

place prior to successful angiogenesis and which then facilitates the vascular, invasive growth. Experimental results have demonstrated that tumour cells secrete both growth-activating and growth-inhibiting chemicals. The models presented in Section 2 have focussed attention on the chemical inhibition of mitosis within multicell spheroids, the main assumption of the modelling being that a growth inhibitory factor (GIF) is produced within the spheroid in some prescribed spatially-dependent manner to reflect the observed cellular heterogeneity within spheroids. As stated in this section, the existence and properties of chemicals which inhibit mitosis are very well documented, e.g., [38–46,77]. However, there is also much evidence to demonstrate that tumour cells also secrete growth-promoting factors [78–80].

We consider then the production of growth promoting factors (GPF) and growth inhibitory factors (GIF) by tumour cells during the avascular phase of growth. The theory will be presented for solid spherical tumours, but the analysis goes over equally well to other similar geometries, e.g., oblate spheroid, prolate spheroid, ellipsoid, etc. It is known, for example, that carcinoma arising in the breast tend to be ellipsoidal in shape. We consider the case whereby the growth factors are produced throughout the interior of the tumour and also the case whereby production of the factors is restricted to the thin layer of live, proliferating cells at the tumour surface. We will discuss the relevance of the resulting spatially inhomogeneous patterns to the experimentally observed cellular heterogeneity within carcinoma and multicell spheroids and also to the well known invasion characteristics of carcinoma.

The system we study is the set of nonlinear partial differential equations which we write in the general form familiar from the theory of reaction-diffusion equations:

$$\frac{\partial \mathbf{c}}{\partial t} = \mathbf{f}(\mathbf{c}) + D\nabla^2\mathbf{c}, \tag{4.1}$$

where $\mathbf{c}$ is the vector of growth factor concentrations, $\mathbf{f}$ represents the reaction kinetics of the chemicals and D is the diagonal matrix of positive constant diffusion coefficients, cf. [81–83]. If we denote the permeability of the tumour tissue at the tumour surface by P, then appropriate boundary conditions for the above system are of the form

$$D(\mathbf{n}.\nabla)\mathbf{c} = -P\mathbf{c}, \qquad \mathbf{r} \text{ on } \partial\Omega, \tag{4.2}$$

where $\partial\Omega$ is the closed boundary of the domain, i.e., the tumour surface, and $\mathbf{n}$ is the outward unit normal. A derivation of these boundary conditions may be found in [31,32] and references therein. Appropriate initial conditions will also be prescribed. For our purposes, we consider only two chemical species whose concentrations are c_1 and c_2, say, representing some growth-promoting factor (GPF) and some growth-inhibiting factor (GIF), respectively. The reaction kinetics governing the system may either be a pure or cross activator-inhibitor mechanism [81,83]. The distinction between these two types of kinetics lies in whether the self-activating chemical either activates (pure) or inhibits (cross) the second species. There is experimental evidence which suggests both mechanisms may be applicable in the case of growth factors secreted by tumour cells [84]. For our purposes throughout this paper, we shall assume that the kinetics are a cross activator-inhibitor mechanism, e.g., the prototypical Schnakenberg system [83]. Under appropriate scaling, it can be shown [81,83] that the above system of equations then reduces to the following:

$$\frac{\partial u}{\partial t} = \gamma f(u,v) + \nabla^2 u, \qquad 0 \le r < 1, \tag{4.3}$$

$$\frac{\partial v}{\partial t} = \gamma g(u,v) + d\nabla^2 v, \qquad 0 \le r < 1, \tag{4.4}$$

where, with R an appropriate tumour radius, k an appropriate reaction parameter and D_1, D_2 the diffusion coefficients of the GPF and GIF, respectively, $\gamma = R^2k/D_1$ and $d = D_2/D_1$. The

boundary conditions for the above system are then given by

$$\frac{\partial u}{\partial r} = -\rho_1 u, \qquad \text{on } r = 1 \tag{4.5}$$

$$\frac{\partial v}{\partial r} = -\rho_2 v, \qquad \text{on } r = 1, \tag{4.6}$$

where $\rho_i = RP/D_i$. The homogeneous steady-state (u_0, v_0) of the above equation is the positive solution of

$$f(u,v) = 0, \qquad g(u,v) = 0. \tag{4.7}$$

Using standard linear stability theory [81,82], it can be shown that the stability of the above steady state is governed by the nature of the eigenvalues λ satisfying the following characteristic equation:

$$\lambda^2 + \lambda\left[\kappa_{nl}^2(1+d) - \gamma\left(f_u + g_v\right)\right] + \left[d\kappa_{nl}^4 - \gamma\left(df_u + g_v\right)\kappa_{nl}^2 + \gamma^2\left(f_u g_v - f_v g_u\right)\right] = 0. \tag{4.8}$$

The steady state will be unstable with respect to spatial disturbances if $\mathrm{Re}(\lambda) < 0$. The parameters κ_{nl} are such that they satisfy the eigenvalue problem

$$\nabla^2\psi + \kappa^2\psi = 0, \qquad \mathbf{r} \in \Omega \tag{4.9}$$

subject to

$$\frac{\partial \psi}{\partial n} + \rho\psi = 0, \qquad \mathbf{r} \text{ on } \partial\Omega. \tag{4.10}$$

Since we are considering the production of the GPF and GIF within a sphere, then the eigenfunctions ψ are given by

$$\psi_{nlm}(r,\theta,\phi) = j_n\left(\kappa_{nl} r\right) Y_{nm}(\theta,\phi), \tag{4.11}$$

where $j_n(\kappa_{nl} r)$ are the spherical Bessel functions and $Y_{nm}(\theta,\phi)$ are the (surface) spherical harmonics. From the boundary condition at $r = 1$, the wavenumbers κ_{nl} satisfy

$$\kappa_{nl} j_n'\left(\kappa_{nl}\right) + \rho j_n\left(\kappa_{nl}\right) = 0. \tag{4.12}$$

It is well known that for certain conditions on the reaction kinetics and $d > 1$ [81,83], there exists a range of unstable wavenumbers $\kappa_1^2 < \kappa^2 < \kappa_2^2$ for which $\mathrm{Re}(\lambda) > 0$. Within this unstable range, $\mathrm{Re}(\lambda)$ will have a maximum which indicates a fastest growing mode which will eventually dominate over time. This will give rise to a spatially heterogeneous pattern.

4.1. Results

From the results of the previous section, we assume that all conditions necessary for diffusion driven instability to occur are satisfied and that there exists a dominant, fastest growing mode. As stated previously, we also assume that the reaction kinetics are of a cross activator-inhibitor mechanism, i.e., regions of high concentration of GPF correspond to regions of low concentration of GIF and vice versa [81]. The following figures illustrate what relevance this could have for the growth of solid tumours. We denote the wavenumber of the fastest growing mode by κ_{nl}^* where from (4.12), we have

$$\kappa_{nl}^* j_n'\left(\kappa_{nl}^*\right) + \rho j_n\left(\kappa_{nl}^*\right) = 0. \tag{4.13}$$

We now give an estimate for the variable $\rho = RP/D$. In [31,32] (and references therein), it was demonstrated that reasonable estimates for a range of values for the diffusion coefficient D and the permeability P were 10^{-6}–$10^{-8}\,\mathrm{cm}^2\,\mathrm{s}^{-1}$ and 10^{-5}–$10^{-6}\,\mathrm{cm}\,\mathrm{s}^{-1}$, respectively. The radius of an avascular multicell spheroid is at most a few millimetres, i.e., of the order 10^{-1}–10^{-2} cm. Hence within this range of parameters, we can choose $\rho = 1$ enabling us to use standard tables

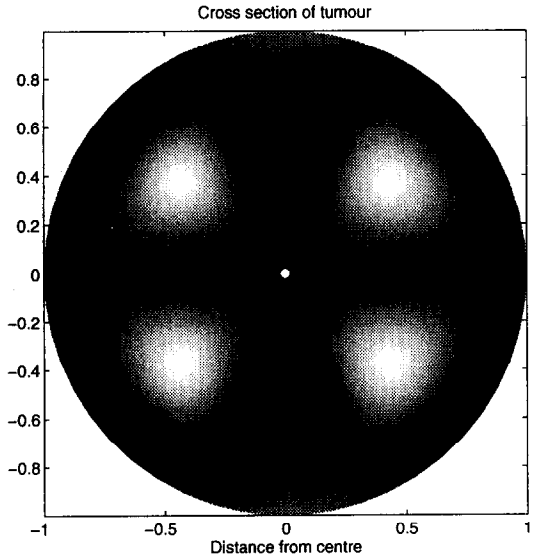

Figure 4.1. A cross section of a solid tumour showing the prepattern generated when the dominant mode is $j_4(\kappa_{42}r)Y_{42}(\theta,\phi)$, with $\kappa_{42} = 9.84$. The light regions are those where the local concentration of a growth promoting factor is high. It is from these regions that recruitment of the quiescent cells into the proliferating compartment is proposed to take place.

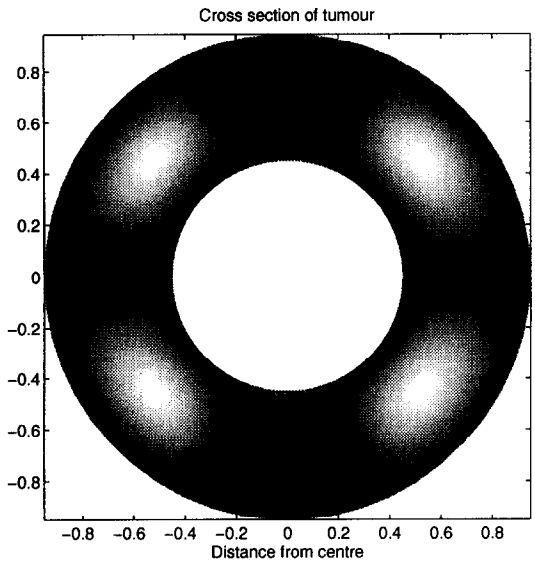

Figure 4.2. A cross section of a solid tumour showing the prepattern generated when the dominant mode is $j_4(\kappa_{42}r)Y_{42}(\theta,\phi)$, with $\kappa_{42} = 9.84$ under the assumption that chemicals are only produced within the quiescent compartment. The light regions are those where the local concentration of a growth promoting factor is high. It is from these regions that recruitment of the quiescent cells into the proliferating compartment is proposed to take place.

to estimate κ^*_{nl}. For illustrative purposes only, Figure 4.1 shows a cross-section of a solid tumour in which the dominant mode prepattern is $j_4(\kappa_{42}r)Y_{42}(\theta,\phi)$, with $\kappa_{42} = 9.84$. This situation is illustrative of growth factors being produced throughout the tumour. As is clearly seen from the figure, there are regions in the interior of the tumour where the concentration of the GPF is high and the GIF low (the light regions). These regions are situated in the area of the interior of the tumour where the quiescent cells are likely to be situated. It is known that these quiescent cells can be recruited to repopulate the proliferating compartment [3,4,19]. From Figure 4.1, we suggest that a possible mechanism for this is that the quiescent cells are first stimulated

into proliferating in those regions of high GPF and are then recruited into the proliferating compartment. A similar situation exists if it is assumed that the chemicals are only produced within the quiescent layer, and this is illustrated in Figure 4.2.

If we now consider growth factors being produced only by the live cells at the surface of the tumour, then the problem is essentially a 2-dimensional one in the variables θ and ϕ. The eigenfunctions are simply the surface harmonics $Y_{nm}(\theta, \phi)$ and the wavenumbers in this case are given by $\kappa_n^2 = n(n+1)$. In this case, zero flux boundary conditions are assumed. Figure 4.3 illustrates the case when the dominant mode is given by $Y_{63}(\theta, \phi)$. As can clearly be seen, once again there are regions on the surface of the tumour where the concentration of the GPF is locally high (dark regions). It is a well-known feature of carcinomas that they invade the surrounding local tissue with columns of cells projecting from the central mass [59]. We suggest that while in its avascular dormant state, a prepattern such as is illustrated in Figure 4.3 is set up. Once vascularized, the cells which are in regions of high GPF will begin to invade the local tissues. A prepattern of this type is also consistent with the observation that tumours can manipulate the local environment by secretion of growth factors. Thus this prepattern will not only prime the tumour cells but will be affecting the local surrounding tissue as well thus facilitating invasion [85].

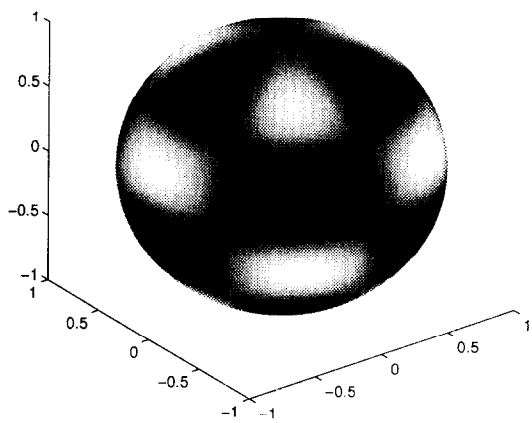

Figure 4.3. Prepattern observed on the surface of a solid tumour when the dominant mode is $Y_{63}(\theta, \phi)$. Once again, the light regions are those in which the concentration of a growth promoting factor is high. It is from these regions that the columns of invading cells are expected to emerge when exophytic growth is taking place. These regions are also expected to influence the local environment through their secretion of growth factors.

5. THE GROWTH AND STABILITY OF A SOLID TUMOUR

The mathematical model of Section 2 provides an informative description of the (implicit) growth of a solid tumour (multicell spheroid) that is essentially spherical in shape, with similar geometrical assumptions holding for the model of Section 4. In this ideal geometry, the net resultant of all intercellular forces within the tumour is quite simple in that it is always radially directed; no more precise examination of local cell dynamics is needed. However, this is not the case for growth under asymmetric conditions or for the displacements produced by nonuniform cellular aggregation. Analysis of these more realistic circumstances requires the specification of the macroscopic laws which govern the jostling movement of crowded cells within an expanding or contracting culture. Although there is still rather little data on this subject, a very reasonable assumption is that gross internal forces may be characterised by a pressure distribution (cf. interstitial pressure), the nonuniformities of which effect cell motion. In addition, cell adhesion, like molecular attraction, is assumed to produce a surface tension at the outer boundary of the colony that maintains compactness and counteracts internal expansive pressures (it is well known that multicell spheroids are held together by a variety of cell–cell junctions including desmosomes, tight junctions, junctional complexes and gap junctions). The quasibalance of these forces causes

what may be termed regular growth or demise of the population. However, a serious imbalance can occur which affects the basic stability and organisation of the culture.

In this section, we present a mathematical model for the growth of a solid tumour due to Chaplain [86] which develops and extends an original model due to Greenspan [48]. Recent work on this "biophysical modelling" approach has also been done by Baxter and Jain [87–89] and Jain [90] with a view to the problem of drug delivery to solid tumours. As we have seen in the introduction, the growth of a tumour is an extremely complex process involving many different variables, and so in order to focus on the main qualitative features of shape and structure, a number of simplifying but realistic approximations are invoked and set out below. These are based on observations from experimental studies on multicell spheroids [4].

- Although the growth of solid tumours in animals always involves some vascularisation (i.e., the existence of a network of blood vessels within the tumour), the earliest stages of development are apparently regulated by the direct diffusion of nutrients and wastes from and to the surrounding tissues. The tumour (or cell culture) and the surrounding tissues (or ambient medium), then, are assumed to be essentially in a state of diffusive equilibrium at all times. The diffusion of oxygen (an indispensable nutrient) over the sizes which characterise the tumour's size and structure is very small compared with the total period of growth. For example, it takes about 10 seconds (1000 seconds) for oxygen to diffuse across a distance of 100μ (1000μ). Both of these time intervals are certainly very short compared to a growth period measured in days, and they may even be regarded as small relative to the twelve or eighteen hour period for mitosis (in epithelial cells of a Chinese hamster). The approximation of instantaneous diffusive equilibrium is based on these disparate time periods.

  When the tumour is small, every cell receives adequate nourishment by simple diffusion. However, the consumption of a nutrient means that its concentration must decrease towards the centre of the tumour. Eventually the concentration there of a vital nutrient falls below the critical level to sustain life and a central necrotic core develops. We thus restrict ourselves to a discussion of tumours in a stage of growth that has a two-layer structure: an outer shell of viable or proliferating cells that envelops a larger core of necrotic debris. Only the proliferating cells consume the nutrient.
- A tumour cell dies when the available concentration of a vital nutrient denoted by $\sigma(x, y, z, t)$ falls below a critical level σ_1. The local thickness h of the thin layer of living cells depends only on σ_1 and the value of σ at the outer surface of the tumour. This relationship is approximated by
$$h = \begin{cases} \nu\sqrt{\sigma - \sigma_1}, & \text{if } \sigma > \sigma_1, \\ 0, & \text{if } \sigma < \sigma_1. \end{cases} \tag{5.1}$$
  This relationship was demonstrated experimentally by Greenspan [55].
- All living tumour cells are identical and each is considered to be an incompressible structure of constant volume. Cell division occurs "instantaneously" relative to the growth time of the tumour, and each daughter cell occupies the same volume as any other cell of the population.
- The mitotic index is constant. If dA is an element of surface area of the tumour, then the incremental volume of live cells, $hdA = dV$, creates a new cell volume at the rate βhdA. Nutrient is consumed by this volume at the rate γhdA.
- There is an adhesion or surface-tension among the living tumour cells, modelling the cell–cell junctional contacts, which, in a spherical geometry, produces an inward pressure that maintains the tumour as a compact, solid and continuous mass. This surface-tension force T is proportional to the mean curvature of the boundary.
- Necrotic cellular debris continually disintegrates into simpler chemical compounds that are freely permeable through cell membranes. The mass, or equivalently, the cell volume

lost in this way in necrosis is replaced by cells pushed inward by the forces of adhesion and surface-tension, and also by stroma and blood vessels. This rate of loss of necrotic mass (or cell volume) per unit volume of debris is taken to be a constant.

- The birth or death of cells produces internal pressure differentials which cause the motion of cellular material. This slow drift is assumed to be governed by

$$\vec{\mathbf{q}} = -\nabla p, \tag{5.2}$$

where $\vec{\mathbf{q}}(x, y, z, t)$ is the particle velocity and $p(x, y, z, t)$ is proportional to the internal pressure.

These last three assumptions constitute an explanation of why cell proliferation can and indeed must continue even when the tumour is in a quiescent steady state. In equilibrium, cells produced in the growth layer flow inward, exactly compensating the loss of cell volume in the necrotic centre. Moreover, a tumour grows when the rate of cell production exceeds the mass and volume loss due to necrotic disintegration. In the reverse situation, the tumour contracts. The loose analogy is to view the tumour as a fluid held together by adhesive forces. Mitosis or cell proliferation acts like a source of incompressible fluid, while necrosis has the role of a fluid sink.

The first objective is to replace the chemical and dynamical processes within the thin layer of viable cells by a set of equivalent conditions that relate variable changes across this region of the tumour. To this end, let the outer surface be represented by

$$\Gamma(x, y, z, t) = 0.$$

Conservation of mass applied to a small volume element on the surface of the spheroid implies that since h is small, the mass/volume flow out of the ends of the element $(\vec{\mathbf{q}}_+ \,.\, \hat{\mathbf{n}} - \vec{\mathbf{q}}_- \,.\, \hat{\mathbf{n}})dA$ is approximately equal to the rate of mass/volume production within this small volume, $\beta h dA$.

Therefore,

$$\vec{\mathbf{q}}_+ \,.\, \hat{\mathbf{n}} = \vec{\mathbf{q}}_- \,.\, \hat{\mathbf{n}} + \beta h. \tag{5.3}$$

Similarly, the rate of nutrient diffusion into dV through the outer surface, $k\hat{\mathbf{n}} \,.\, \nabla\sigma dA$, is equal to the rate at which nutrient is consumed in this small volume, $\gamma h(\sigma)dA$, that is

$$k\hat{\mathbf{n}} \,.\, \nabla\sigma = \gamma h(\sigma). \tag{5.4}$$

(Since $\sigma = \sigma_1$ in the necrotic core, there is no diffusive transport from the interior.) Here the convection of the nutrient across the surface is assumed to be negligible compared to that transported by diffusion in an equal time.

Furthermore, we assume that the proliferation rate is large so that its product with the very small thickness results in an order one quantity, i.e.,

$$\beta h = \beta\nu\sqrt{\sigma - \sigma_1} = \lambda\sqrt{\sigma - \sigma_1}, \qquad \text{with } \lambda = 0(1).$$

Similarly,

$$\gamma h = \gamma\nu\sqrt{\sigma - \sigma_1} = \mu k\sqrt{\sigma - \sigma_1}, \qquad \text{with } \mu = 0(1).$$

To this order of accuracy, the pressure and the tangential velocity components are continuous across the stratum of live cells:

$$p_+ = p_-, \qquad \vec{\mathbf{q}}_+ \times \hat{\mathbf{n}} = \vec{\mathbf{q}}_- \times \hat{\mathbf{n}}.$$

The pressure on the surface of the tumour must equal the surface tension force, and if κ_1 and κ_2 are the principal curvatures of the boundary, then

$$p = \frac{\alpha}{2}(\kappa_1 + \kappa_2) = \alpha\kappa, \tag{5.5}$$

where α is a constant and κ is the mean curvature of the boundary.

The preceding conditions apply at the moving interface $\Gamma(x, y, z, t) = 0$, and the equation of motion of a point on this surface is

$$\frac{d\vec{\mathbf{r}}}{dt} = \vec{\mathbf{q}}_+. \tag{5.6}$$

The initial geometry of the tumour, of course, must be prescribed.

In the core where the rate of volume loss per unit volume is a constant, S_i, the expression of mass conservation is

$$\nabla.\vec{\mathbf{q}} = -S_i. \tag{5.7}$$

The equation for the nutrient concentration σ, which is assumed to be in diffusive equilibrium, is, outside the colony

$$\nabla^2\sigma = 0, \tag{5.8}$$

whereas in the necrotic interior,

$$\sigma = \sigma_1. \tag{5.9}$$

A source of nutrient in the finite domain adds an inhomogeneous term to the right-hand side of (5.8). The formulation is completed by specifying the manner in which the tumour is supplied nourishment.

Two interesting problems on the time dependent response of a tumour that can be examined with the theory as formulated are:

- growth with a constant supply of nutrient at infinity, i.e., $\sigma \to \sigma_\infty$ as $|\vec{\mathbf{r}}| \to \infty$;
- the effect of a nearby source of nutrient.

By using (5.2) in conjunction with (5.7), and hence eliminating the velocity vector from the problem, the mathematical formulation can be given entirely in terms of the scalar functions p and σ. A convenient form for the first boundary value problem to be studied retains interior pressure and the velocity of the tumour surface:

$$\nabla^2 p = S_i, \qquad \text{inside } \Gamma = 0, \tag{5.10}$$

$$\nabla^2\sigma = 0, \qquad \text{outside } \Gamma = 0, \tag{5.11}$$

where on $\Gamma(x, y, z, t) = 0$,

$$\begin{aligned} p &= \frac{\alpha}{2}(\kappa_1 + \kappa_2) = \alpha\kappa, \\ \vec{\mathbf{q}}_+.\hat{\mathbf{n}} &= -\hat{\mathbf{n}}.\nabla p + \lambda\sqrt{\sigma - \sigma_1}, \\ \vec{\mathbf{q}}_+ \times \hat{\mathbf{n}} &= -\nabla p \times \hat{\mathbf{n}}, \\ \hat{\mathbf{n}}.\nabla\sigma &= \mu\sqrt{\sigma - \sigma_1}. \end{aligned} \tag{5.12}$$

The bounding surface is defined by

$$\frac{d\vec{\mathbf{r}}}{dt} = \vec{\mathbf{q}}_+, \tag{5.13}$$

and the initial parametric prescription

$$\vec{\mathbf{r}} = \vec{\mathbf{a}}(\xi,\ \zeta) \qquad \text{at } t = 0. \tag{5.14}$$

5.1. The Initial Growth Phase

Under conditions of complete spherical symmetry, all cell variables depend only on r and t, and the tumour grows as a sphere of radius $R(t)$. The boundary problem summarised in (5.10)–(5.13) then reduces to

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial p}{\partial r}\right) = S_i, \qquad r \le R(t), \tag{5.15}$$

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial \sigma}{\partial r}\right) = 0, \qquad r \ge R(t), \tag{5.16}$$

with boundary conditions

$$\sigma \to \sigma_\infty \qquad \text{as } r \to \infty, \tag{5.17}$$

and

$$\begin{aligned} p &= \frac{\alpha}{R(t)}, \\ R'(t) &= -\frac{\partial p}{\partial r} + \lambda\sqrt{\sigma - \sigma_1}, \\ \frac{\partial \sigma}{\partial r} &= \mu\sqrt{\sigma - \sigma_1}, \end{aligned} \tag{5.18}$$

on $r = R(t)$ and $R(0) = a$.

The third equation in the boundary conditions (5.12), i.e., $\vec{\mathbf{q}}_+ \times \hat{\mathbf{n}} = -\nabla p \times \hat{\mathbf{n}}$, is trivially satisfied since $\vec{\mathbf{q}}_+ = (R'(t), 0, 0)$, $\nabla p = (\frac{\partial p}{\partial r}, 0, 0)$, $\hat{\mathbf{n}} = (1, 0, 0)$ and both vector products give $\mathbf{0}$

It is convenient to use the relative nutrient concentration $\sigma - \sigma_1$ as a basic variable, and this can be accomplished either through a variable change or by setting $\sigma_1 = 0$. The latter convention will be used from now on.

Since the time variable enters the system only in determination of the tumour radius, the preceding system of equations can be solved explicitly for p and σ:

$$p = \frac{S_i}{6}\left(r^2 - R^2\right) + \frac{\alpha}{R}, \tag{5.19}$$

$$\sigma = \sigma_\infty - \left(\frac{2\mu\sigma_\infty R^2}{\mu R + \sqrt{\mu^2 R^2 + 4\sigma_\infty}}\right)\frac{1}{r}. \tag{5.20}$$

The substitution of these formulæ into (5.18) yields a single ordinary differential equation for $R(t)$:

$$R'(t) = -\frac{1}{3}S_i R + \frac{2\lambda\sigma_\infty}{\mu R + \sqrt{\mu^2 R^2 + 4\sigma_\infty}}, \tag{5.21}$$

with $R(0) = a$. The last equation can then be integrated to yield an unwieldy closed form solution which will not be quoted here since only the qualitative character of $R(t)$ is of interest.

The tumour (multicell spheroid) grows initially at a linear rate and asymptotes to the steady state radius

$$R_\infty = \frac{3\lambda}{S_i}\left(\frac{\sigma_\infty}{1 + 3\lambda\mu/S_i}\right)^{1/2}, \tag{5.22}$$

in a time scale regulated mainly by the rate of volume loss of necrotic debris (this can be obtained by setting the right hand side of (5.12) to zero).

The surface tension which maintains the compactness of the tumour has an otherwise implicit role in the development of the growth of the spherical tumour. However, it is of crucial importance in determining whether such symmetrical growth is indeed stable and hence attainable.

Growth is judged unstable to infinitesimal perturbations if any such disturbance amplifies at an exponential rate (exceeding that of the radius $R(t)$). In this circumstance, the instabilities radically alter the shape of the colony and can even lead to fracturing into two or more pieces. The tumour becomes unstable if and when it reaches a critical size beyond which surface tension is overcome by pressure forces. The precise criterion for this is obtained from the analysis that follows.

We now consider again the growth of a spherical tumour which is subjected to small deviations that are always inevitably present in a real environment. Originally, for simplicity, Greenspan [48] took the perturbations from complete sphericity to be axially symmetric, i.e., independent of the azimuthal angle ϕ in spherical polar coordinates (r, θ, ϕ). Here, however, for completeness, we will include an azimuthal dependence in our perturbations, cf. [86].

To this end, we let $\overline{p}(r,t)$, $\overline{\sigma}(r,t)$, and $R(t)$ denote the basic state of motion given by equations (5.19)–(5.21) and $\epsilon\tilde{p}(r,\theta,\phi,t)$, $\epsilon\tilde{\sigma}(r,\theta,\phi,t)$, and $\epsilon\xi(\theta,\phi,t)$ the perturbations therefrom. The total pressure distribution and nutrient concentration are then represented by

$$\begin{aligned} p(r,\theta,\phi,t) &= \overline{p}(r,t) + \epsilon\tilde{p}(r,\theta,\phi,t) \\ \sigma(r,\theta,\phi,t) &= \overline{\sigma}(r,t) + \epsilon\tilde{\sigma}(r,\theta,\phi,t) \end{aligned} \tag{5.23}$$

and the equation of the moving surface is now given by

$$\Gamma(r,\theta,\phi,t) = r - R(t) - \epsilon\xi(\theta,\phi,t) = 0. \tag{5.24}$$

These expressions are substituted into (5.10)–(5.23) and all terms are developed as power series in ϵ, including the mean curvature and the unit normal vector of the distorted surface. This application of classical perturbation theory, precise details of which are given in [86], immediately yields (5.15)–(5.18), as well as the equations governing the deviations from the basic evolutionary state. It follows that

$$\nabla^2\tilde{p} = 0, \qquad r \le R(t), \tag{5.25}$$

$$\nabla^2\tilde{\sigma} = 0, \qquad r \ge R(t), \tag{5.26}$$

with $\tilde{\sigma} \to 0$ as $r \to \infty$.

While on $r = R(t)$, we have

$$\frac{\partial\xi}{\partial t} = -\left(\frac{\partial^2\overline{p}}{\partial r^2}\xi + \frac{\partial\tilde{p}}{\partial r}\right) + \frac{\lambda}{2\sqrt{\overline{\sigma}}}\left(\xi\frac{\partial\overline{\sigma}}{\partial r} + \tilde{\sigma}\right), \tag{5.27}$$

$$\frac{\partial^2\overline{\sigma}}{\partial r^2}\xi + \frac{\partial\tilde{\sigma}}{\partial r} = \frac{\mu}{2\sqrt{\overline{\sigma}}}\left(\xi\frac{\partial\overline{\sigma}}{\partial r} + \tilde{\sigma}\right), \tag{5.28}$$

$$\frac{\partial\overline{p}}{\partial r}\xi + \tilde{p} = -\frac{\alpha}{2R^2}\left[\frac{\partial}{\partial\eta}\left((1-\eta^2)\frac{\partial\xi}{\partial\eta}\right) + 2\xi + \frac{1}{\sin^2\theta}\frac{\partial^2\xi}{\partial\phi^2}\right], \tag{5.29}$$

where

$$\eta = \cos\theta.$$

Modal solutions of this homogeneous system are given by

$$\tilde{p} = A_n(t)r^nP_n^m(\cos\theta)\cos m\phi, \tag{5.30}$$

$$\tilde{\sigma} = B_n(t)P_n^m(\cos\theta)\frac{\cos m\phi}{r^{n+1}}, \tag{5.31}$$

$$\xi = C_n(t)P_n^m(\cos\theta)\cos m\phi, \tag{5.32}$$

where $n = 0,1,2,3,\ldots$, $0 \le m \le n$ and $P_n^m(\cos\theta)$ are the associated Legendre polynomials defined by

$$P_n^m(\cos\theta) = \sin^m\theta\frac{d^m}{d(\cos\theta)^m}\left[P_n(\cos\theta)\right]$$

(in the case $m = 0$, these reduce to the Legendre polynomials $P_n(\cos\theta)$ and the whole analysis corresponds to [48] where there is no azimuthal dependence).

Also, we have from the boundary equations, the following relations:

$$A_n(t) = -\frac{C_n(t)}{R^n}\left[\overline{p}_r - \frac{\alpha}{2R^2}(n+2)(n+1)\right]_{r=R}, \tag{5.33}$$

$$B_n(t) = R^{n+2}C_n(t)\left[\frac{\overline{\sigma}_{rr}\sqrt{\overline{\sigma}} - (\mu/2)\overline{\sigma}_r}{(n+1)\sqrt{\overline{\sigma}} + \mu/(2R)}\right]_{r=R}, \tag{5.34}$$

$$\frac{dC_n(t)}{dt} = C_n(t)\left[-\overline{p}_{rr} + \frac{n}{R}\overline{p}_r - \frac{\alpha n}{2R^3}(n-1)(n+2) + \frac{\lambda}{2}\left(\frac{(n+1)\overline{\sigma}_r + R\overline{\sigma}_{rr}}{(n+1)\sqrt{\overline{\sigma}} + \mu R/2}\right)\right]_{r=R}. \tag{5.35}$$

The result of replacing $\overline{p}$, $\overline{\sigma}$ and their derivatives by explicit expressions obtained from equations (5.19) and (5.20) is the fundamental equation governing the growth of infinitesimal instabilities

$$C_n'(t) = (n-1)f(n,R)C_n(t), \tag{5.36}$$

where

$$f(n,R) = \frac{S_i}{3} - \frac{\alpha n(n+2)}{2R^3} + \frac{(\lambda\mu/2)}{(n+1) + (\mu R/4\sigma_\infty)\left[\mu R + (\mu^2R^2 + 4\sigma_\infty)\right]^{1/2}}. \tag{5.37}$$

The disturbance amplifies or decays according to whether the product $(n-1)f(n,R)$ is positive or negative. The tumour development is unstable if small perturbations can amplify—otherwise it is stable. For $n = 0$, the time dependent coefficient in $-f(0,R)$ in equation (5.36) is always negative so that this mode is definitely stable. For $n = 1$, $C_1(t)$ is a constant, which is compatible with the fact that this mode represents a translation of axes.

The modes $n \geq 2$ are genuinely unstable when $f(n,R) > 0$. Moreover, since $f(n,R)$ is a decreasing function of the index n for $n \geq 1$, the mode $n = 2$ is the first that can become unstable and it must always be a component of any growing disturbance. The minimum critical radius R_c at which the developing tumour can become unstable is then given by

$$f(2,R_c) = 0. \tag{5.38}$$

If $R_c > R_\infty$, the steady state radius of the tumour (see (5.22)), then the tumour is stable during its entire growth and attains its symmetric equilibrium configuration.

If, however, $R_c < R_\infty$, then the tumour becomes unstable at some definite time in its growth when small disturbances amplify and change both the structure and the shape of the tumour. In this case, the changing pressure distribution overcomes the surface tension before spherical equilibrium is reached.

The onset of stability in the mode $n = 2$ is manifested as a pinch in the outer surface around the equatorial region. As the tumour grows and increases in size, other modes become unstable and more radical changes in configuration may occur, which can be used to model the tumour invading the surrounding tissues. This process is described in the following section.

5.2. Asymmetric Growth, Local Invasion and Metastasis

Clinically, tumours can be classified into two main types—benign and malignant—although these two terms really represent the extremes of a spectrum of tumour growth. We now wish to show that the predictions of, and the parameters involved in, the fundamental equations governing the growth of infinitesimal instabilities—(5.36),(5.37)—and also the modes of instability arising from these equations, describe very well the characteristics of, and the situations which arise in, malignant, benign and those tumours of an intermediate nature. We turn first of all to malignant tumours (i.e., cancers) and in particular to the type which makes up the great majority (90%) of malignant tumours, carcinomas [Greek: *Karkinos*, a crab].

At the margin of a carcinoma, single tumour cells and irregular columns or projections of cells infiltrate into the surrounding tissues, extending from the central mass like the legs of a crab. As they proliferate, the invading cells compress the tissue cells and interfere with their blood supply, and highly specialised cells such as the parenchymal cells of internal organs are destroyed. Fibrovascular tissue is more resistant and its growth is stimulated by the tumour cells which thus secure a supporting stroma and blood supply.

Growth occurs most readily along lines of least resistance such as planes of loose connective tissue. Denser tissue, for example fibrous fasciae, the walls of arteries, cartilage and compact bone, are relatively resistant to invasion, and while this is largely explained by their structure,

there is evidence that constituent pieces of cartilage can both interfere with the growth of cancer cells and inhibit angiogenesis.

Local invasion is very important, not only because it increases the difficulty of excising a carcinoma completely, but also for its effects. In particular, invasion in and around the walls of hollow viscera or gland ducts very commonly causes their obstruction, while involvement of nerves may cause pain. The sooner a carcinoma gives rise to local effects of this sort, the greater the chance of diagnosis and complete removal.

This invasive power of cancers may be attributed to three main factors:

- the abnormal motility and amoeboid movement of some cancer cells,
- the secretion of lytic enzymes, e.g., hyaluronidase which dissolve the ground substance and digest formed tissue elements and damage cell membranes,
- the reduced adhesiveness of cancer cells to one another.

When considered alongside two other important features of malignant tumours, i.e., a large rate of cell production and extensive necrosis, we see from (5.37) that high modes of instability are possible.

We recall:

$$f(n,R) = \frac{S_i}{3} - \frac{\alpha n(n+2)}{2R^3} + \frac{(\lambda\mu/2)}{(n+1) + (\mu R/4\sigma_\infty)\left[\mu R + (\mu^2R^2 + 4\sigma_\infty)\right]^{1/2}}.$$

Thus the combination of a high value for S_i (extensive necrosis), a high value for λ (fast proliferation rate), and a small value for the surface tension parameter α (reduced cell-to-cell adhesiveness) leads to the possibility of an instability being dominated by a high mode number n. The mode n that actually dominates the instability is selected by the function $R(t)$ which sets the amplification rate and the time available for each perturbation to amplify, while the selection of the m-mode (cf. $P_n^m(\cos\theta)$) will be determined by external influences such as the denseness of the surrounding tissue and the external distribution of the enzymes secreted by the tumour. Surface energy considerations may also play a part, and these will be discussed in the next section along with cell mobility. Final domination by a mode such as $P_6^3(\cos\theta)$ leads to the typical crab-like appearance assumed by carcinomas as illustrated in Figure 5.1.

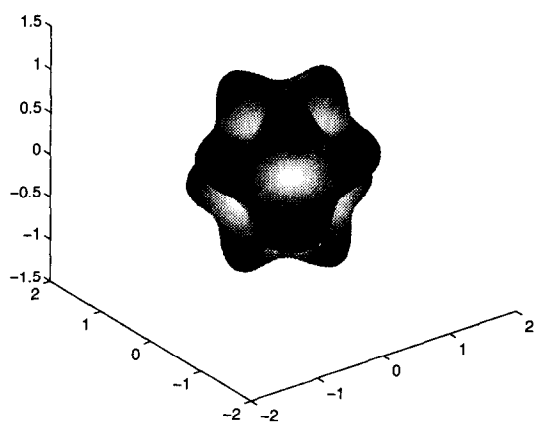

Figure 5.1. Development of an instability in an invading carcinoma showing classical exophytic growth pattern of columns of cancer cells projecting into external tissue.

At the other end of the spectrum, we have benign tumours. Generally these show little or no necrosis, have a slower rate of cell proliferation than malignant tumours, and are very often surrounded by a fibrous capsule. Even when no such capsule exists, the margin between the tumour and the surrounding tissues remains sharp, without evidence of local invasion. Again, upon consideration of the function $f(n,R)$, this time with a low value for S_i (little sign of necrosis), a low value for λ (slower cell proliferation rate), and a high value for α the surface-tension

parameter (modelling the effect of the fibrous capsule), we see that the possibility of domination by a high mode number n is very small. Indeed, the presence of the capsule, which keeps the tumour a compact spherical mass, effectively rules out the growth of any small instability except perhaps the mode $n = 0$, which is simply associated with radial growth. This models the noninvasive nature of benign tumours.

For tumours of an intermediate nature, i.e., combining some of the properties of both malignant and benign tumours, the choice of mode number n will not be as clear cut, depending upon a more complicated interplay between the parameters involved and the external influences. This will vary upon the degree of malignancy from tumour to tumour.

As well as invading surrounding tissues, tumour cells also invade the walls of lymphatics and blood vessels in and around the tumour. This gives rise to the possible spread of the tumour to other parts of the body where secondary tumours or metastases then form. We examine this process of metastasis in the light of the abnormal motility of some cancer cells and the reduced cell adhesiveness using the third equation of (5.12).

Smaller lymphatics, whose walls consist of little more than an endothelium with loose intercellular junctions and an incomplete basement membrane, are readily invaded by carcinoma cells which may either become detached and carried in the lymph to the next draining lymph node, or they may remain attached at the site of invasion and proliferate within the lymphatic to form a continuous column of tumour which distends and obstructs the lumen.

Carcinoma cells also invade the walls of small vessels, particularly venules, in and around the tumour, and single cancer cells or small groups of cells frequently become detached and are carried away in the venous blood where they are likely to become impacted as emboli in the next microvascular network.

There is clinical and experimental evidence that rough handling of a malignant tumour promotes the escape of tumour cells into the blood and increases the risk of metastasis. Like the primary tumour, metastases are locally invasive and their cells also enter the lymphatics and blood vessels and cause further dissemination by these routes.

Using the third equation of (5.12), i.e.,

$$\vec{\mathbf{q}}_+ \times \hat{\mathbf{n}} = -\nabla p \times \hat{\mathbf{n}}$$

enables us to follow the locus of a particular cell as it moves about on the surface of the tumour. Initially, since all variables depend only upon r and t, the velocity of a particular cell has a radial component only, i.e., $\vec{\mathbf{q}}_+ = (R'(t), 0, 0)$. Upon consideration of the introduction of small perturbations, we assume that each cell now has tangential components of velocity, describing its possible motion in both the θ- and ϕ-directions. From equation (5.24), and assuming that the new angular velocity components are of first order in ϵ, we thus have

$$\vec{\mathbf{q}}_+ = \left(R'(t) + \epsilon\xi_t,\ \epsilon R\dot{\theta},\ \epsilon R\dot{\phi}\sin\theta\right), \tag{5.39}$$

where the dots denote differentiation w.r.t. time.

Calculation of the vector products to order ϵ then gives us the following equations for the angular components of velocity:

$$\begin{aligned} R\dot{\phi}\sin^2\theta &= -\left[\frac{2S_i}{3} + \frac{R'}{R}\right]\xi_\phi - \frac{\tilde{p}_\phi}{R}, \\ R\dot{\theta} &= -\left[\frac{2S_i}{3} + \frac{R'}{R}\right]\xi_\theta - \frac{\tilde{p}_\theta}{R}. \end{aligned} \tag{5.40}$$

For highly malignant tumours, we saw that it was possible for an instability to be dominated by a high mode number n. It is also known that certain malignant tumour cells display an abnormal degree of cell motility. Thus we can see from (5.40) that if we have a dominant high mode number,

then this allows for the cells on the surface of the tumour to have fairly substantial movement in both the θ- and the ϕ-directions and so to move about quite freely over the tumour surface. When this is viewed in the light of the fact that malignant tumours have a greatly reduced cell-to-cell adherence and also that the surface tension distribution over the tumour surface is given by

$$\frac{\alpha}{R} + \frac{\epsilon\alpha C_n(t)}{2R^2}(n-1)(n+2)P_n^m(\cos\theta)\cos m\phi, \tag{5.41}$$

we see that a cell or groups of cells may come across areas of the tumour surface where the surface tension is reduced, and so there is the possibility that single cells or groups of cells may become detached and carried away in the bloodstream, as previously described, leading to the formation of secondary tumours elsewhere in the body. Again, the more malignant the tumour is, the more mobile cells will be on the surface and so the more likely metastasis is to occur.

5.3. Selection of the Mode of Instability

Since the onset of instability in a malignant tumour and its subsequent invasion of the surrounding tissues is characterised by a process involving a certain furrowing of the tumour surface and the projection of crab-like legs of tumour cells from the central mass outwards into the tissue, we attempt to describe this invasiveness partly as the result of a minimisation procedure on a function which in some way represents a generalised surface energy. The invasion of the surrounding tissue by a malignant tumour can then be viewed partly as a series of transformations to successively higher energy levels. To this end, we assume that there exists some field function u defined over the surface of the tumour such that its nodal lines represent the lines of least resistance to a furrowing or "pinching" process manifested in the onset of instability of the tumour. In our case, it is assumed that u is some sort of order-disorder parameter relating to the organisation of the live, proliferating cells on the tumour surface. It is reasonable to suppose [86,91,92] in problems of this nature that the surface energy density is given by an expression of the form

$$\mathcal{E}(\theta,\ \phi) = D\left[\left(\frac{\partial u}{\partial\theta}\right)^2 + \frac{1}{\sin^2\theta}\left(\frac{\partial u}{\partial\phi}\right)^2 + \beta u^2\right], \tag{5.42}$$

where D and β are constants incorporating the physiological properties of the tumour surface (e.g., cell-to-cell adhesion, cell mobility, etc.). The total surface energy E is then given by

$$E = \int_{dA} \mathcal{E}(\theta,\ \phi)\, dA. \tag{5.43}$$

We now suppose that the onset of instability corresponds to a minimum of E, i.e.,

$$\delta E = \delta \int_0^{2\pi}\int_0^{\pi} \mathcal{E}(\theta,\ \phi)\sin\theta\, d\theta\, d\phi = 0, \tag{5.44}$$

subject to a conservation law on u^2

$$\int_0^{2\pi}\int_0^{\pi} u^2(\theta,\ \phi)\sin\theta\, d\theta\, d\phi = 1. \tag{5.45}$$

Equations (5.44),(5.45) are then required to satisfy the Euler-Lagrange equation

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial u}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 u}{\partial\phi^2} - \gamma u = 0, \tag{5.46}$$

where γ incorporates the parameter β and an undetermined multiplier of (5.45).

With the conditions that u must be finite, single-valued and continuous over the tumour, this leads to the solution

$$u(\theta,\ \phi) = \sqrt{2}N_{mn}P_n^m(\cos\theta)\cos m\phi, \tag{5.47}$$

where N_{mn} are normalisation constants and the parameter γ takes the value $n(n+1)$. From (5.43), we can now easily calculate the surface energy corresponding to each mode number n. Thus we have

$$E_n = D[n(n+1) + \beta]. \tag{5.48}$$

However, not every characteristic state of (5.48) is realised in the growth and invasion process, since there are many external influences which impose restrictions on the choice of mode number n. The degree of malignancy of the tumour will also play a part. For highly malignant tumours whose cells have reduced adhesion to one another and greater mobility, coupled with the fact that secretion of enzymes by the tumour will affect the surrounding tissues, we envisage the possibility of dominance by a high mode number n which gives the tumour a sufficiently large surface energy E to invade the surrounding tissues extensively.

5.4. Modification of the Nutrient Supply

We now consider the behaviour of a tumour which has reached its steady state radius R_∞ (and as such is in a stable state of equilibrium) to a slight modification in its supply of nutrient. We can model this by placing a weak source or sink of nutrient somewhere in the surrounding tissue. The reaction of a tumour to a nearby sink models the interaction of two neighbouring tumours in competition for the same nutrient supply, while the reaction of a tumour to a nearby source could be taken to model the effect of extra nutrient supply derived from neighbouring blood vessels after the secretion of Tumour Angiogenesis Factor. In order to account for a weak source placed at, say, $x = 0$, $y = 0$, $z = l$, then we write (5.8) as

$$\nabla^2\sigma = \epsilon\sigma_\infty R_\infty \delta(x)\delta(y)\delta(z-l)H(t), \tag{5.49}$$

where $H(t)$ is the Heaviside Function defined by

$$H(t) = \begin{cases} 1, & t > 0, \\ 0, & t < 0. \end{cases}$$

All the remaining equations and boundary conditions are the same, with the exception that we now assume that the tumour is initially in a stable state of equilibrium, i.e., a sphere of radius R_∞. We again use perturbation theory in our analysis, replacing $R(t)$ by R_∞ and equation (5.26) by

$$\nabla^2\tilde{\sigma} = \sigma_\infty R_\infty \delta(x)\delta(y)\delta(z-l)H(t). \tag{5.50}$$

Now the inhomogeneous system (5.25),(5.50),(5.27)–(5.29) is solved with initial conditions $\tilde{p} = 0$, $\tilde{\sigma} = 0$, $\xi = 0$. Assuming axial symmetry, the solution to the above system is given by

$$\tilde{p} = \sum_0^\infty A_n(t)P_n(\cos\theta)r^n, \tag{5.51}$$

$$\tilde{\sigma} = \frac{\sigma_\infty R_\infty H(t)}{4\pi\left(r^2 - 2rl\cos\theta + l^2\right)^{1/2}} + \sum_0^\infty \frac{B_n(t)P_n(\cos\theta)}{r^{n+1}}, \tag{5.52}$$

$$\xi = \sum_0^\infty C_n(t)P_n(\cos\theta). \tag{5.53}$$

Using the well-known expansion

$$\left(r^2 - 2rl\cos\theta + l^2\right)^{-1/2} = \sum_0^\infty \frac{r^n}{l^{n+1}}P_n(\cos\theta) \qquad \text{for } r \le l, \tag{5.54}$$

and then substituting equations (5.51)–(5.53) into (5.27)–(5.29) leads to the inhomogeneous counterparts of (5.33)–(5.35), for $n = 0, 1, 2, \ldots$. Now for the mode $n = 1$, representing a linear translation, we have

$$C_1(t) = \frac{R_\infty^2 \lambda}{16\pi l^2}\left[\mu R_\infty + \left(\mu^2 R_\infty^2 + 4\sigma_\infty\right)^{1/2}\right] t + C_1(0). \tag{5.55}$$

Thus we can see that the tumour begins to move towards the nutrient source with a velocity that is inversely proportional to the square of the distance l, and we note that Gimbrone *et al.* [59] report the movement of tumour implants towards the limbal vessels in their angiogenesis experiments.

6. DISCUSSION AND CONCLUSIONS

In this paper, we have presented a variety of mathematical models describing the various stages of the growth and development of a solid tumour.

In Section 2, we focussed attention upon the production of a growth inhibitory factor (GIF) within a spherical avascular nodule (multicell spheroid) and showed that using a nonlinear, spatially-dependent diffusion coefficient, which reflects the cellular heterogeneity of the interior of multicell spheroids and the heterogeneous intercellular permeability, is alone sufficient to produce a GIF concentration profile within the spheroid which is not incompatible with available experimental evidence. Similar results were also obtained by using a combination of nonlinear diffusion with a nonlinear source term, and hence this combination can also account for the experimental observations. These results show that a nonuniform source function is not the only way to produce qualitatively correct GIF concentration profiles within multicell spheroids. Therefore, mathematically it is not possible to distinguish between the effects of nonlinear diffusion and a nonlinear source term. We conclude that more experimental work on the precise mechanisms governing cell cycle kinetics must be done to elucidate the underlying tissue heterogeneity.

In Section 3, a mathematical model for tumour angiogenesis was presented which developed and extended the model of Chaplain and Stuart [61]. As is apparent from its brief description in the introduction, the complete process of angiogenesis is a complicated one, involving several distinct, and not necessarily related, events. This in turn requires several separate mechanisms which can stimulate each event—e.g., vasodilation, endothelial cell migration, endothelial cell proliferation and loop formation (anastomosis), to name a few. Despite much research and many advances, many questions still remain unanswered [25]. To formulate a single mathematical model which would include all of these processes would be very difficult indeed. In this paper, we have chosen to focus attention primarily on the activity of the endothelial cells, since they are always at the heart of the angiogenic process, as well as on the TAF concentration profile. We have modelled in a simple but effective manner two of the three main events concerning the endothelial cells (migration and proliferation), and the results obtained are in good agreement with both experimental evidence and other models [25,27,63] and represent an improvement upon those obtained by Chaplain and Stuart [61]. The model reproduces the experimentally observed facts that the initial response of the endothelial cells is one of migration with proliferation occurring as a secondary response which is nevertheless vital for the successful completion of angiogenesis.

The results also confirm previous assumptions and findings that the endothelial cell migration is almost certainly controlled by some form of taxis rather than diffusion. However, since endothelial cells are known to exhibit different levels of motility on different extracellular molecules [93–95], it would certainly be possible to modify the model to account for this variable motility. More specifically we could assume a more general form of (3.10), namely

$$\mathbf{J}_{\text{diffusion}} = -D_n(c)\nabla n. \tag{6.1}$$

The diffusion, or motility, coefficient of the cells would now be dependent upon the concentration of the TAF. This means considering a nonlinear diffusion problem.

The extension of the model to 2 spatial dimensions was made in order to take into account explicitly the key features of anastomosis and secondary sprout formation (branching). The numerical simulations show that the basic assumptions of the original model can capture most of the salient features in 2 spatial dimensions with only slight modifications necessary. However, some further points of clarification remain. It was found difficult to simulate two capillary sprout tips moving towards each other (in opposite directions) and fusing together. The features captured by the model are on a rather coarser level. This is perhaps to be expected since certain simplifying assumptions have been made and certain biological details omitted.

Although we have chosen to focus on chemotaxis as the underlying driving mechanism since there is much experimental evidence to support this, there is also experimental evidence that the interaction between the endothelial cells and the extracellular matrix, through which they must move to reach the tumour, may also have a part to play in the angiogenic process. It is known for instance that the EC secrete chemicals which may give rise to haptotaxis (motion up a gradient of increased adhesiveness). From *in vitro* experiments, it is known that the mobilization of endothelial cells can be greatly enhanced by adding gangliosides to whatever medium they are cultivated upon. It has been shown that preincubating endothelial cells with trisialogangliosides and then culturing on the same substratum leads to their *binding* to fibronectin being greatly enhanced and their migration is increased five-fold [25,94,96]. Including haptotaxis within the model would not be difficult to do from a modelling point of view but would increase the computational time for the simulations drastically since several chemicals would perhaps need to be considered. Also we note that in simulating the brush-border effect, we have not prescribed the precise *mechanism* which gives rise to this event, only assumed that it occurs and incorporated appropriate term in the equation which accounts for this. Investigation of precise mechanisms which give rise to the brush-border and also to the side-branching is a possible line of further research. It may be that a combination of deterministic and stochastic modelling is necessary.

In Section 4, we applied standard reaction-diffusion theory (Turing-type models) to a novel situation—that of the growth of solid tumours, e.g., carcinoma. We have shown that the spatially heterogeneous patterns which arise in the case of a spherical geometry may play a part and help to explain certain observed phenomenon in carcinoma and multicell spheroids, i.e., the characteristic invasive pattern and the recruitment of quiescent cells into the proliferating population. We note that similar results can be obtained for other quasi-spherical geometries which may also be appropriate for solid tumours, e.g., oblate spheroid, prolate spheroid, ellipsoid, cf. [97,98]. Of course there are many other factors which are involved in tumour growth of this type, e.g., nutrient supply, and which are also very important, and we certainly do not claim that Turing systems can provide a complete answer in this particular case, but rather may be an important part of the complex overall mechanisms governing solid tumour growth.

We note that there are several features of solid tumour growth which lend themselves naturally to the consideration of a Turing-type model. There are actually present in this system chemicals which both promote and inhibit growth of cells. The time scale for the growth of a solid tumour is very slow in comparison to the time needed for a diffusing substance to reach steady state concentration, and hence, there will be a genuine prepattern. There also appears to be a natural critical domain size in this system, i.e., the size of carcinoma in its diffusion-limited, avascular state. While in this state, no invasion can take place, but once vascularized, rapid exophytic growth does occur.

We also note that many of the standard problems inherent in Turing models regarding robustness, i.e., the same pattern must be repeated faithfully time and time again with the requirement that only a small number of patterns are selected in a robust and controlled manner, are not present for solid tumour growth. The observed "invasion patterns" of *in vivo* carcinoma vary greatly from individual to individual. It is also well known that cellular heterogeneity is not only observed within an individual tumour/multicell spheroid [3,19] but also varies from tumour to tumour. Also it is worth pointing out that the the complex process of invasion is dependent upon

many factors such as the actual make up of the surrounding tissue. In view of the latter point regarding the surrounding tissue, this may well necessitate a reconsideration of the boundary conditions involved which may lead to even more complex patterning [81].

In Section 5, the study of the asymmetric development of a compact, spherical tumour growing initially by expansion was presented, based on several simplifying assumptions about the macroscopic nature of the intercellular forces. A surface-tension force proportional to the mean curvature of the tumour surface is assumed to maintain the tumour as a compact and continuous mass, consisting of a thin layer of live cells undergoing mitosis surrounding a necrotic core. As new cells form in the outer mitotic layer and expand to their proper size, they must push aside neighbouring cells of the tumour. These forces of displacement are transmitted with attenuation, cell-to-cell, throughout the crowded population. The total pressure in this way causes the internal migration of cells and a drift of the entire tumour as it builds in the direction of a richer nutrient supply. A "seepage" law of motion is adopted in which internal motion is proportional to the negative gradient of pressure. In essence, the model constructed may be compared to the motion of an incompressible fluid within a closed, extensible membrane which has a distribution of sources and sinks that depend on time and other factors.

It was shown that under certain conditions, the tendency of the tumour to distort by either growth or the elimination of material in the necrotic core overcomes the stabilising effect of surface tension. The tumour development is then unstable, and it may achieve a nonlinear equilibrium, pinch apart or extensively invade the surrounding tissues, depending upon the dominant modes of instability and their development in time. These possibilities are linked to the degree of malignancy of the tumour, with more malignant tumours being more likely to show asymmetrical development because of certain factors inherent in their make-up. Most notably, a greatly reduced cell-to-cell adhesion, which diminishes the surface-tension force holding the tumour together as a compact mass, an increased cell mobility, allowing individual cells and groups of cells to move freely over the tumour surface with the possibility that upon coming across an area of the tumour surface of reduced surface-tension, the cells may become detached and carried away in the bloodstream to form metastases in distant parts of the body. The effect of the secondary perturbations introduced in the model, as well as accounting for the deviations in pressure and nutrient supply, may also be taken to simulate the rough handling of tumours which is known to cause the detachment of cells. By attributing a surface energy E to the live cells, it was shown that the process of invasion by a malignant tumour may be thought of in part as a transition from a low energy level (that of the initial spherical mass) to a high energy level (that of the invading malignancy), with the final dominant mode number n being sufficiently high to allow the tumour to overcome any repressive external influence, such as dense surrounding tissue, and to assume the typical crab-like shape. Recent work by Chaplain and Sleeman [99] using nonlinear elasticity theory also attempts to demonstrate this from a biophysical viewpoint. Finally, we demonstrated that a tumour which has developed to its stable, steady state equilibrium can react both to a change in its nutrient supply by moving towards the extra nutrient supply. This may well model the response of a tumour to neovascularisation, as reported by Gimrone *et al.* [59].

In each of the models presented in this paper, there are strengths and weaknesses. Since each different model focuses on different aspects of solid tumour growth and makes the relevant appropriate simplifying assumptions, this is to be expected. However, it is to be hoped that in the model of the final chapter a certain coming together of the models of Sections 2–4 is to be seen, and it is certainly to be hoped that in future, mathematical models of solid tumour growth will develop so as to be of genuine prognostic use to the pathologist.

REFERENCES

1. J. Folkman, Tumor angiogenesis, *Cancer* **3**, 355–388 (1975).
2. J. Folkman, The vascularization of tumors, *Sci. Am.* **234**, 58–73 (1976).

3. R.E. Durand, Multicell spheroids as a model for cell kinetic studies, *Cell Tissue Kinet.* **23**, 141–159 (1990).
4. R.M. Sutherland, Cell and environment interactions in tumor microregions: The multicell spheroid model, *Science* **240**, 177–184 (1988).
5. W. Mueller-Klieser, Multicellular spheroids: A review on cellular aggregates in cancer research, *J. Cancer Res. Clin. Oncol.* **113**, 101–122 (1987).
6. F. Brümmer and D.F. Hülser, Intercellular communication in multicell spheroids, *European J. Cell Biol.* **27**, 108 (1982).
7. H. Dertinger, G. Hinz and K.H. Jakobs, Intercellular communication, three-dimensional cell contact and radiosensitivity, *Biophys. Struct. Mech.* **9**, 89–93 (1982).
8. G. Adam, U. Steiner, H. Maier and S. Ullrich, Analysis of cellular interactions in density-dependent inhibition of 3T3 cell proliferation, *Biophys. Struct. Mech.* **9**, 75–82 (1982).
9. R.E. Durand, Cell cycle kinetics in an *in vitro* tumour model, *Cell Tissue Kinet.* **9**, 403–412 (1976).
10. J.P. Freyer and R.M. Sutherland, Determination of diffusion constants for metabolites in multicell tumor spheroids, *Adv. Exp. Med. Biol.* **159**, 463–475 (1983).
11. J.P. Freyer, Role of necrosis in saturation of spheroid growth, *Strahlentherapie* **160**, 58 (1984).
12. J.P. Freyer, E. Tustanoff, A.J. Franko and R.M. Sutherland, *In situ* oxygen consumption rates of cells in V-79 multicellular spheroids during growth, *J. Cell. Physiol.* **118**, 53–61 (1984).
13. J.P. Freyer and R.M. Sutherland, Regulation of growth saturation and development of necrosis in EMT6/Ro multicellular spheroids by the glucose and oxygen supply, *Cancer Res.* **46**, 3504–3512 (1986).
14. J.P. Freyer and R.M. Sutherland, Proliferative and clonogenic heterogeneity of cells from EMT6/Ro multicellular spheroids induced by the glucose and oxygen supply, *Cancer Res.* **46**, 3513–3520 (1986).
15. J.P. Freyer and P.L. Schor, Regrowth of cells from multicell tumour spheroids, *Cell Tissue Kinet.* **20**, 249 (1987).
16. R.M. Sutherland, J.A. McCredie and W.R. Inch, Growth of multicell spheroids as a model of nodular carcinomas, *J. Nat. Cancer Inst.* **46**, 113–120 (1971).
17. R.M. Sutherland and R.E. Durand, Growth and cellular characteristics of multicell spheroids, *Recent Results in Cancer Research* **95**, 24–49 (1984).
18. P.W. Vaupel, S. Frinak and H.I. Bicher, Heterogeneous oxygen partial pressure and pH distribution in C3H mouse mammary adenocarcinoma, *Canc. Res.* **41**, 2008–2013 (1981).
19. E. Wibe, T. Lindmo and O. Kaalhus, Cell kinetic characteristics in different parts of multicellular spheroids of human origin, *Cell Tissue Kinet.* **14**, 639–651 (1981).
20. K. Groebe and W. Mueller-Klieser, Distributions of oxygen, nutrient and metabolic waste concentrations in multicellular spheroids and their dependence on spheroid parameters, *Eur. Biophys. J.* **19**, 169–181 (1991).
21. B. Bourrat-Floeck, K. Groebe and W. Mueller-Klieser, Biological response of multicellular EMT6 spheroids to exogenous lactate, *Int. J. Cancer* **47**, 792–799 (1991).
22. A. Bredel-Geissler, U. Karbach, S. Walenta, L. Vollrath and W. Mueller-Klieser, Proliferation-associated oxygen consumption and morphology of tumor cells in monolayer and spheroid culture, *J. Cell. Physiol.* **153**, 44–52 (1992).
23. J.M. Brown and A.J. Gaccia, Tumour hypoxia: The picture has changed in the 1990s, *Int. J. Radiat. Biol.* **65**, 95–102 (1994).
24. J. Folkman and M. Klagsbrun, Angiogenic factors, *Science* **235**, 442–447 (1987).
25. N. Paweletz and M. Knierim, Tumor-related angiogenesis, *Crit. Rev. Oncol. Hematol.* **9**, 197–242 (1989).
26. S. Paku and N. Paweletz, First steps of tumor-related angiogenesis, *Lab. Invest.* **65**, 334–346 (1991).
27. M.M. Sholley, G.P. Ferguson, H.R. Seibel, J.L. Montour and J.D. Wilson, Mechanisms of neovascularization. Vascular sprouting can occur without proliferation of endothelial cells, *Lab. Invest.* **51**, 624–634 (1984).
28. J.A. Adam, A simplified mathematical model of tumor growth, *Math. Biosci.* **81**, 229–244 (1986).
20. J.A. Adam, A mathematical model of tumor growth II: Effects of geometry and spatial nonuniformity on stability, *Math. Biosci.* **86**, 183–211 (1987).
30. J.A. Adam, A mathematical model of tumor growth III: Comparison with experiment, *Math. Biosci.* **86**, 213–227 (1987).
31. M.A.J. Chaplain, D.L. Benson and P.K. Maini, Nonlinear diffusion of a growth inhibitory factor in multicell spheroids, *Math. Biosci.* **121**, 1–13 (1994).
32. M.A.J. Chaplain and N.F. Britton, On the concentration profile of a growth inhibitory factor in multicell spheroids, *Math. Biosci.* **115**, 233–245 (1993).
33. J. Landry, J.P. Freyer and R.M. Sutherland, A model for the growth of multicell spheroids, *Cell Tissue Kinet.* **15**, 585–594 (1982).
34. R.M. Shymko and L. Glass, Cellular and geometric control of tissue growth and mitotic instability, *J. Theor. Biol.* **63**, 355–374 (1976).
35. G.W. Swan, The diffusion of an inhibitor in a spherical tumor, *Math. Biosci.* **108**, 75–79 (1992).
36. J.J. Casciari, S.V. Sotirchos and R.M. Sutherland, Mathematical modelling of microenvironment and growth in EMT6/Ro multicellular tumour spheroids, *Cell Prolif.* **25**, 1–22 (1992).
37. F.L. Degner and R.M. Sutherland, Mathematical modelling of oxygen supply and oxygenation in tumor tissues: Prognostic, therapeutic and experimental implications, *Int. J. Radiation Oncology Biol. Phys.* **15**, 391–397 (1988).
38. G. Brugal and J. Pelmont, Existence of two chalone-like substances in intestinal extract from the adult newt inhibiting embryonic intestinal cell proliferation, *Cell Tiss. Kinet.* **8**, 171–187 (1975).

39. W. Hondius-Boldingh and E.B. Laurence, Extraction, purification and preliminary characterization of the epidermal chalone, *J. Biochem.* **5**, 191–198 (1968).
40. O.H. Iversen, Epidermal chalones and squamous cell carcinomas, *Virchows Arch. B Cell Path.* **27**, 229–235 (1978).
41. O.H. Iversen, The chalones, In *Tissue Growth Factors*, (Edited by R. Baserga), pp. 491–550, Springer-Verlag, Berlin, (1981).
42. O.H. Iversen, What's new in endogenous growth stimulators and inhibitors (chalones), *Path. Res. Pract.* **180**, 77–80 (1985).
43. O.H. Iversen, The hunt for endogenous growth-inhibitory and or tumor suppression factors—Their role in physiological and pathological growth-regulation, *Adv. Cancer Res.* **57**, 413–453 (1991).
44. F.A. Marks, Tissue-specific factor inhibiting DNA synthesis in mouse epidermis, *Natn. Cancer Inst. Monogr.* **38**, 79–90 (1973).
45. A.E. Levine, D.A. Hamilton, L.C. Yeoman, H. Busch and M.G. Brattain, Identification of a tumor inhibitory factor in rat ascites fluid, *Biochem. Biophys. Res. Commun.* **119**, 76–82 (1984).
46. L. Harel, G. Chatelain and A. Golde, Density-dependent inhibition of growth: Inhibitory diffusible factors from 3T3—and Rous sarcoma virus (RSV)—transformed 3T3 cells, *J. Cell Physiology* **119**, 101–106 (1984).
47. N.F. Britton and M.A.J. Chaplain, A qualitative analysis of some models of tissue growth, *Math. Biosci.* **113**, 77–89 (1993).
48. H.P. Greenspan, On the growth and stability of cell cultures and solid tumours, *J. Theor. Biol.* **56**, 229–242 (1976).
49. W.R. Loewenstein, Junctional intercellular communication: The cell-to-cell membrane channel, *Physiol. Rev.* **61**, 829–889 (1981).
50. D.F. Hülser and F. Brümmer, Closing and opening of gap junction pores between two- and three-dimensionally cultured tumor cells, *Biophys. Struct. Mech.* **9**, 83–88 (1982).
51. U. Karbach, C-D. Gerharz, K. Groebe, H.E. Gabbert and W. Mueller-Klieser, Rhabdomyosarcoma spheroids with central proliferation and differentiation, *Cancer Res.* **52**, 474–477 (1992).
52. W.S. Bullough and J.U. Deol, The pattern of tumour growth, *Symp. Soc. Exp. Biol.* **25**, 255–275 (1971).
53. A.C. Burton, Rate of growth of solid tumours as a problem of diffusion, *Growth* **3**, 157–176 (1966).
54. H.P. Greenspan, Models for the growth of a solid tumor by diffusion, *Stud. Appl. Math.* **51**, 317–340 (1972).
55. H.P. Greenspan, On the self-inhibited growth of cell cultures, *Growth* **38**, 81–95 (1974).
56. D.L.S. McElwain and P.J. Ponzo, A model for the growth of a solid tumor with non-uniform oxygen consumption, *Math. Biosci.* **35**, 267–279 (1977).
57. S.A. Maggelakis and J.A. Adam, Mathematical model of prevascular growth of a spherical carcinoma, *Mathl. Comput. Modelling* **13** (5), 23–38 (1990).
58. J. Folkman and M. Hochberg, Self-regulation of growth in three dimensions, *J. Exp. Med.* **138**, 745–753 (1973).
59. M.A. Gimbrone, R.S. Cotran, S.B. Leapman and J. Folkman, Tumor growth and neovascularization: An experimental model using the rabbit cornea, *J. Natn. Cancer Inst.* **52**, 413–427 (1974).
60. V.R. Muthukkaruppan, L. Kubai and R. Auerbach, Tumor-induced neovascularization in the mouse eye, *J. Natn. Cancer Inst.* **69**, 699–705 (1982).
61. M.A.J. Chaplain and A.M. Stuart, A model mechanism for the chemotactic response of endothelial cells to tumour angiogenesis factor, *IMA J. Math. Appl. Med. Biol.* **10**, 149–168 (1993).
62. D.H. Ausprunk and J. Folkman, Migration and proliferation of endothelial cells in preformed and newly formed blood vessels during tumour angiogenesis, *Microvasc. Res.* **14**, 53–65 (1977).
63. C.L. Stokes and D.A. Lauffenburger, Analysis of the roles of microvessel endothelial cell random motility and chemotaxis in angiogenesis, *J. Theor. Biol.* **152**, 377–403 (1991).
64. D. Balding and D.L.S. McElwain, A mathematical model of tumour-induced capillary growth, *J. Theor. Biol.* **114**, 53–73 (1985).
65. M.A.J. Chaplain and A.M. Stuart, A mathematical model for the diffusion of tumour angiogenesis factor into the surrounding host tissue, *IMA J. Math. Appl. Med. Biol.* **8**, 191–220 (1991).
66. J.A. Sherratt and J.D. Murray, Models of epidermal wound healing, *Proc. R. Soc. Lond. B* **241**, 29–36 (1990).
67. W.J. Cliff, Observations on healing tissue: A combined light and electron microscopic investigation, *Phil. Trans. R. Soc. Lond. B* **246**, 305–325 (1963).
68. G.I. Schoefl, Studies on inflammation III. Growing capillaries: Their structure and permeability, *Virchows Arch. Pathol. Anat.* **337**, 97–141 (1963).
69. B.A. Warren, The growth of the blood supply to melanoma transplants in the hamster cheek pouch, *Lab. Invest.* **15**, 464–473 (1966).
70. L.A. Liotta, G.M. Saidel and J. Kleinerman, Diffusion model of tumor vascularization and growth, *Bull. Math. Biol.* **39**, 117–129 (1977).
71. M.A.J. Chaplain and B.D. Sleeman, A mathematical model for the production and secretion of tumour angiogenesis factor in tumours, *IMA J. Math. Appl. Med. Biol.* **7**, 93–108 (1990).
72. N.A. Wright, Cell proliferation kinetics of the epidermis, In *Biochemistry and Physiology of the Skin*, (Edited by L.A. Goldsmith), pp. 203–229, Oxford University Press, (1987).
73. M.A. Reidy and S.M. Schwartz, Endothelial regeneration III. Time course of intimal changes after small defined injury to rat aortic endothelium, *Lab. Invest.* **44**, 301–312 (1981).

74. C.L. Stokes, M.A. Rupnick, S.K. Williams and D.A. Lauffenburger, Chemotaxis of human microvessel endothelial cells in response to acidic fibroblast growth factor, *Lab. Invest.* **63**, 657–668 (1990).
75. G.D. Winter, Epidermal regeneration studied in the domestic pig, In *Epidermal Wound Healing*, (Edited by H.I. Maibach and D.T. Rovee), pp. 71–112, Chicago Year Book Medical Publishers, (1972).
76. S.K. Williams, Isolation and culture of microvessel and large-vessel endothelial cells: Their use and transport in clinical studies, In *Microvascular Perfusion and Transport in Health and Disease*, (Edited by P. McDonagh), pp. 204–245, Karger, Basle, (1987).
77. K. Parkinson and A. Balmain, Chalones revisited—A possible role for transforming growth-factor-beta in tumor promotion, *Carcinogenesis* **11**, 195–198 (1990).
78. T.C. Chen, D.R. Hinton, M.L.J. Apuzzo and F.M. Hofman, Differential effects of tumor-necrosis-factor-alpha on proliferation, cell-surface antigen expression and cytokine interactions in malignant gliomas, *Neurosurgery* **32**, 85–94 (1993).
79. E. Kawase, H. Yamamoto, K. Hashimoto and N. Nakatsuji, Tumor-necrosis-factor-alpha (TNF-alpha) stimulates proliferation of mouse primordial germ-cells in culture, *Develop. Biol.* **161**, 91–95 (1994).
80. S. Wu, C.M. Boyer, R.S. Whitaker, A. Berchuck, J.R. Wiener, J.B. Wienberg and R.C. Bast, Tumor-necrosis-factor-alpha as an autocrine and paracrine growth-factor for ovarian cancer—Monokine induction of tumor-cell proliferation and tumor-necrosis-factor-alpha expression, *Cancer Res.* **53**, 1939–1944 (1993).
81. R. Dillon, P.K. Maini and H.G. Othmer, Pattern formation in generalized Turing systems. I. Steady-state patterns in systems with mixed boundary conditions, *J. Math. Biol.* **32**, 345–393 (1994).
82. A. Hunding, Dissipative structures in reaction-diffusion systems: Numerical determination of bifurcations in the sphere, *J. Chem. Phys.* **72**, 5241–5248 (1980).
83. J.D. Murray, Parameter space for Turing instability in reaction diffusion mechanisms: A comparison of models, *J. Theor. Biol.* **98**, 143–163 (1982).
84. S. Michelson and J. Leith, Autocrine and paracrine growth factors in tumour growth: A mathematical model, *Bull. Math. Biol.* **53**, 639–656 (1991).
85. J. Keski-Oja, A.E. Postlethwaite and H.L. Moses, Transforming growth factors and the regulation of malignant cell growth and invasion, *Cancer Invest.* **6**, 705–724 (1988).
86. M.A.J. Chaplain, The development of a spatial pattern in a model for cancer growth, In *Experimental and Theoretical Advances in Biological Pattern Formation*, (Edited by H.G. Othmer, P.K. Maini and J.D. Murray), pp. 45–60, Plenum Press, (1993).
87. L.T. Baxter and R.K. Jain, Transport of fluid and macromolecules in tumors: I. Role of interstitial pressure and convection, *Microvasc. Res.* **37**, 77–104 (1989).
88. L.T. Baxter and R.K. Jain, Transport of fluid and macromolecules in tumors: II. Role of heterogeneous perfusion and lymphatics, *Microvasc. Res.* **40**, 246–263 (1990).
89. L.T. Baxter and R.K. Jain, Transport of fluid and macromolecules in tumors: IV. A microscopic model of the perivascular distribution, *Microvasc. Res.* **41**, 252–272 (1991).
90. R.K. Jain, Barriers to drug delivery in solid tumors, *Sci. Amer.* **271**, 58–65 (1994).
91. B.C. Goodwin and L.E.H. Trainor, A field description of the cleavage process in embryogenesis, *J. Theor. Biol.* **86**, 757–770 (1980).
92. L.D. Landau and E.M. Lifschitz, *Theory of Elasticity*, Pergamon Press, London, (1959).
93. V.P. Terranova, R. Diflorio, R.M. Lyall, S. Hic, R. Friesel and T. Maciag, Human endothelial cells are chemotactic to endothelial cell growth factor and heparin, *J. Cell Biol.* **101**, 2330–2334 (1985).
94. S. Ungari, R.S. Katari, G. Alessandri and P.M. Gullino, Cooperation between fibronectin and heparin in the mobilization of capillary endothelium, *Invas. Metast.* **5**, 193–205 (1985).
95. W.C. Young and I.M. Herman, Extracellular matrix modulation of endothelial cell shape and motility following injury *in vitro*, *J. Cell. Sci.* **73**, 19–32 (1985).
96. G. Alessandri, K.S. Raju and P.M. Gullino, Interaction of gangliosides with fibronectin in the mobilization of capillary endothelium, *Invas. Metast.* **6**, 145–165 (1986).
97. A. Hunding, Bifurcations of nonlinear reaction-diffusion systems in prolate spheroids, *J. Math. Biol.* **17**, 223–239 (1983).
98. A. Hunding, Bifurcations of nonlinear reaction-diffusion systems in oblate spheroids, *J. Math. Biol.* **19**, 249–263 (1984).
99. M.A.J. Chaplain and B.D. Sleeman, Modelling the growth of solid tumours and incorporating a method for their classification using nonlinear elasticity theory, *J. Math. Biol.* **31**, 431–479 (1993).
100. M.M. Sholley, M.A. Gimbrone and R.S. Cotran, Cellular migration and replication in endothelial regeneration, *Lab. Invest.* **36**, 18–25 (1977).